Path integral quantization for the gauge field of quantum electrodynamics

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Abstract

The canonical quantization of quantum fields is usually the primary technique that one learns when studying quantum field theory for the first time. It is relatively straightforward for scalar, vector, and even spinor fields, yet becomes arguably complex for gauge fields because of their inherent internal redundancy. This paper provides a detailed and complete derivation of the path integral quantization for the gauge field of quantum electrodynamics. We start with the fundamental concept of Green functions, and then derive the path integral formulation of quantum mechanics. The path integral approach to quantum fields is presented next, followed by its application to the Klein-Gordon scalar field. Finally, the last section shows how to derive the photon two-point correlation function from the electromagnetic field generating functional.

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1 Green functions and propagators

Green functions were first introduced as a method to solve inhomogeneous differential equations [1]. The idea is to find the impulse response of the system which is then used to build a general solution. To illustrate this concept, we will consider a one-dimensional forced harmonic oscillator whose evolution is given by:

$$\left(\frac{d^2}{dt^2} + \omega^2\right) x(t) = J(t) \tag{1}$$

where $x(t)$ is the position at time t, ω is the oscillator's natural frequency, and $J(t)$ is the source. The Green function $G(t, t')$ associated with this system satisfies the following equation:

$$-\left(\frac{d^2}{dt^2} + \omega^2\right) G(t, t') = \delta(t - t') \tag{2}$$

$$G(t, t') = -\left(\frac{d^2}{dt^2} + \omega^2\right)^{-1} \delta(t - t') \tag{3}$$

where $\delta(t - t')$ is Dirac's delta function such that [2]:

$$\int_{-\infty}^{+\infty} \delta(x - a) f(x) dx = f(a) \tag{4}$$

Physically, the Green function $G(t, t')$ represents the system's impulse response to a source J perfectly localized at $t = t'$. If the Green function is known, the general solution to equation (1) is given by:

$$x(t) = -\int_{-\infty}^{+\infty} G(t, t') J(t')\, dt' \tag{5}$$

This is easily shown by substituting this solution into equation (1):

$$\begin{aligned}
x(t) &= \int_{-\infty}^{+\infty} \left(\frac{d^2}{dt^2} + \omega^2\right)^{-1} \delta(t - t') J(t')\, dt' \\
\left(\frac{d^2}{dt^2} + \omega^2\right) x(t) &= \int_{-\infty}^{+\infty} \left(\frac{d^2}{dt^2} + \omega^2\right) \left(\frac{d^2}{dt^2} + \omega^2\right)^{-1} \delta(t - t') J(t')\, dt' \\
\left(\frac{d^2}{dt^2} + \omega^2\right) x(t) &= \int_{-\infty}^{+\infty} \delta(t - t') J(t')\, dt' \\
\left(\frac{d^2}{dt^2} + \omega^2\right) x(t) &= J(t)
\end{aligned}$$

■

Looking at equation (5), one sees why Green functions are also called *propagators*: starting from an impulse disturbance at t', $G(t, t')$ *propagates* its effect on the system at time t. The integral in (5) sums over all the impulse disturbances associated with the time-dependent source $J(t)$ to reconstruct the state of the system at time t.
We now show that the Green function of a harmonic oscillator with natural frequency ω is:

$$G(t, t') = \frac{1}{2\pi} \int \frac{e^{-i(t-t')p}}{p^2 - \omega^2} dp \tag{6}$$

At this stage the dummy integration variable p doesn't mean much, but we will eventually see that it is associated with momentum. We can check the veracity of this claim with a few lines

of calculus:

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = -\left(\frac{d^2}{dt^2}+\omega^2\right)\left(\frac{1}{2\pi}\int \frac{e^{-i(t-t')p}}{p^2-\omega^2}dp\right)$$

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = -\frac{1}{2\pi}\left(\int \frac{\omega^2 e^{-i(t-t')p}}{p^2-\omega^2}dp + \frac{d^2}{dt^2}\int \frac{e^{-i(t-t')p}}{p^2-\omega^2}dp\right)$$

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = -\frac{1}{2\pi}\left(\int \frac{\omega^2 e^{-i(t-t')p}}{p^2-\omega^2}dp + \int \frac{(-ip)^2 e^{-i(t-t')p}}{p^2-\omega^2}dp\right)$$

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = \frac{1}{2\pi}\int \frac{(p^2-\omega^2)e^{-i(t-t')p}}{p^2-\omega^2}dp$$

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = \frac{1}{2\pi}\int e^{-i(t-t')p}dp$$

$$-\left(\frac{d^2}{dt^2}+\omega^2\right)G(t,t') = \delta(t-t')$$

where in the last line we have used the integral form of the delta function to find the definition of the Green function given in (2). The expression for $G(t,t')$ given in equation (6) contains two real poles at $p=\pm\omega$. These singularities can avoided by choosing an appropriate integration contour in the complex plane. We shall focus on three integration contours of particular interest in quantum field theory.

1.1 Retarded propagator

Let $z\in\mathbb{C}$ such that $z=p+i\alpha$ with $\alpha\in\mathbb{R}$, and consider the following contour integral in the complex plane:

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\oint \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = \lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\oint \frac{e^{-i(t-t')z}}{(z+i\varepsilon+\omega)(z+i\varepsilon-\omega)}\,dz \tag{7}$$

The poles have been infinitesimally pushed down to the lower half of the complex plane at $z_1=-\omega-i\varepsilon$ and $z_2=\omega-i\varepsilon$, as shown in Figure 1.

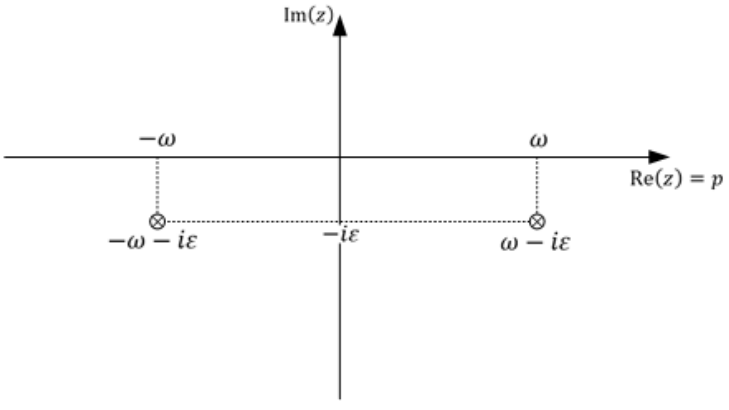


Figure 1: Retarded pole placement

When $t'>t$ we choose the infinite integration contour C_1 plotted in red on Figure 2.
The integration along C_1 consists of

1. Integration on the real axis from $-\infty$ to $+\infty$, which corresponds to the integral we set out to calculate in equation (6).

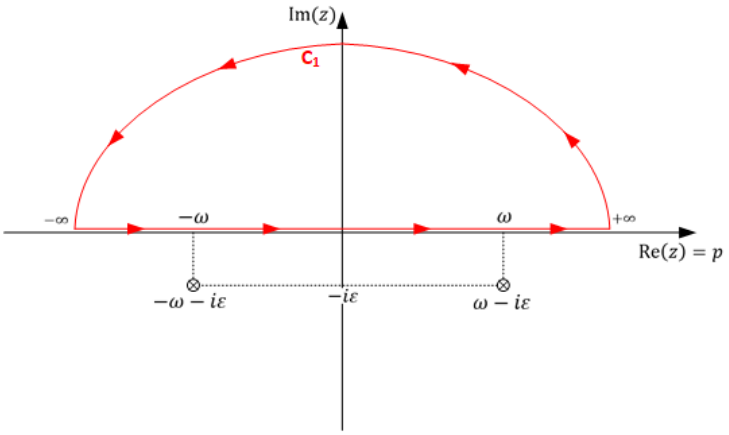


Figure 2: Integration contour C_1 for the retarded propagator

2. Integration, counter-clockwise, along the half-circle of infinite radius located on the upper-half of the complex plane.

Written explicitly, and omitting the limit operator and $1/2\pi$ for clarity, this C_1 contour integral is:

$$\oint_{C_1} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp + \int_{uhhc} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz \tag{8}$$

where *uhhc* denotes the upper-half half-circle of infinite radius. Jordan's lemma states that the integral

$$\int_{uhhc} f(z)e^{ikz}dz$$

vanishes for $k > 0$, provided that $\lim_{M\to\infty} f(Me^{i\theta}) = 0$ [3]. The last integral in equation (8) is therefore zero since we are presently considering the case where $t' > t \Rightarrow -(t-t') > 0$, and the condition $\lim_{M\to\infty} \frac{1}{(Me^{i\theta}+i\varepsilon)^2-\omega^2} = 0$ is satisfied. Equation (8) thus simplifies to:

$$\oint_{C_1} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp \tag{9}$$

The contour integration in (9) can be calculated explicitly using Cauchy's residue theorem [4]: let $f(z)$ be a complex-valued function, and C a closed contour. If $f(z)$ does not have any poles located inside C then

$$\oint_C f(z)dz = 0$$

Otherwise

$$\oint_C f(z)dz = \pm 2\pi i \sum_{z_i} \mathrm{Res}(f(z), z_i)$$

where for a simple pole z_i the residue is defined by $\mathrm{Res}(f(z), z_i) = \lim_{z\to z_i}(z - z_i)f(z)$. The $\pm$ depends on the contour direction: $+$ for counter-clockwise direction, and $-$ for clockwise direction. Since the function $\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}$ does not have poles inside C_1 (cf. Figure 2), Cauchy's residue theorem tells us that

$$\oint_{C_1} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = 0$$

$$\Rightarrow \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = 0 \quad (\text{for } t' > t) \tag{10}$$

We now consider the case $t > t'$ for which we choose the integration contour C_2 depicted in

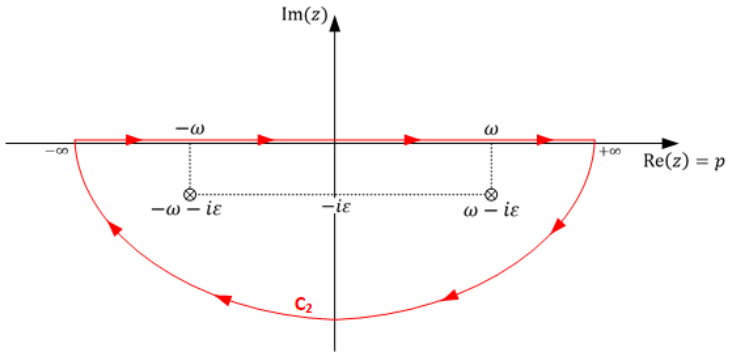


Figure 3: Integration contour C_2 for the retarded propagator

Figure 3.
The C_2 integration contour is given by

$$\oint_{C_2} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp + \int_{lhhc} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz \tag{11}$$

where *lhhc* denotes the lower-half half-circle of infinite radius. Thanks to Jordan's lemma we have

$$\int_{lhhc} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = 0$$

because this time we have $t > t' \Rightarrow -(t-t') < 0$ and we are working in the lower half of the complex plane. Equation (11) thus becomes:

$$\begin{gathered}\oint_{C_2} \frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}\,dz = \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp \\ \Rightarrow \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = \\ -2\pi i\left[\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, -\omega-i\varepsilon\right) + \mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, \omega-i\varepsilon\right)\right]\end{gathered} \tag{12}$$

where in the last equation we have applied Cauchy's residue theorem for the clockwise contour C_2 which contains two poles at $z_1 = -\omega - i\varepsilon$ and $z_2 = +\omega - i\varepsilon$ (cf. Figure 3). We must now calculate the two residues.

$$\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, -\omega-i\varepsilon\right) = \lim_{z\to-\omega-i\varepsilon}(z-(-\omega-i\varepsilon))\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}$$

$$\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, -\omega-i\varepsilon\right) = \lim_{z\to-\omega-i\varepsilon}(z+\omega+i\varepsilon)\frac{e^{-i(t-t')z}}{(z+i\varepsilon+\omega)(z+i\varepsilon-\omega)}$$

$$\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, -\omega-i\varepsilon\right) = \frac{e^{-i(t-t')(-\omega-i\varepsilon)}}{-\omega-i\varepsilon+i\varepsilon-\omega} = \frac{-e^{i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}$$

Similar calculation for the residue at $z_2 = +\omega - i\varepsilon$ yields:

$$\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z+i\varepsilon)^2-\omega^2}, \omega-i\varepsilon\right) = \frac{e^{-i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}$$

After substituting these residues in equation (12) we get:

$$\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = -2\pi i\left(\frac{-e^{i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}+\frac{e^{-i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}\right)$$

Now is the perfect time to reintroduce the limit $\varepsilon \to 0^+$ and the $1/2\pi$ factor we dropped back in equation (8):

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = -i\left(\frac{-e^{i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}+\frac{e^{-i\omega(t-t')}e^{-\varepsilon(t-t')}}{2\omega}\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = \frac{i}{2\omega}\left(e^{i\omega(t-t')}-e^{-i\omega(t-t')}\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = \frac{i}{2\omega}2i\sin\left(\omega(t-t')\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = \frac{-\sin\left(\omega(t-t')\right)}{\omega} \quad (\text{for } t>t') \tag{13}$$

where in the third line we have used Euler's formula $e^{i\alpha}-e^{-i\alpha}=2i\sin\alpha$. Combining equations (10) and (13) gives us:

$$G_+(t'\to t) = \lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{(p+i\varepsilon)^2-\omega^2}\,dp = \begin{cases} -\sin\left(\omega(t-t')\right)/\omega & \text{for } t>t' \\ 0 & \text{for } t'>t \end{cases} \tag{14}$$

where $G_+(t'\to t)$ represents the *retarded propagator* from t' to t. Because it is zero for $t'>t$, it is sometimes referred to as the *causal propagator*.

1.2 Advanced propagator

In this section we repeat the same calculations with a different choice of integration contour for equation (6):

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\oint \frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}\,dz = \lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\oint \frac{e^{-i(t-t')z}}{(z-i\varepsilon+\omega)(z-i\varepsilon-\omega)}\,dz \tag{15}$$

The poles are now infinitesimally pushed to the upper half of the complex plane at $z_1=-\omega+i\varepsilon$ and $z_2=\omega+i\varepsilon$ (cf. Figure 4).

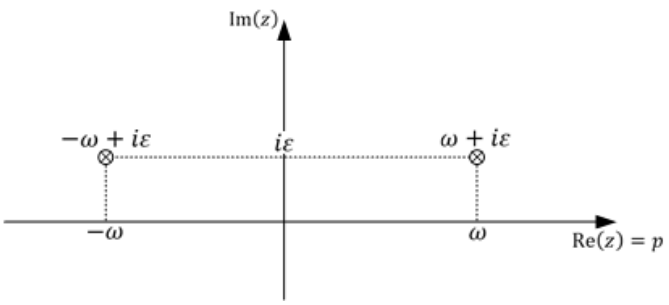


Figure 4: Advanced pole placement

We once again choose contour C_1 for the case $t'>t$, although in this case it encloses the two poles (cf. Figure 5).

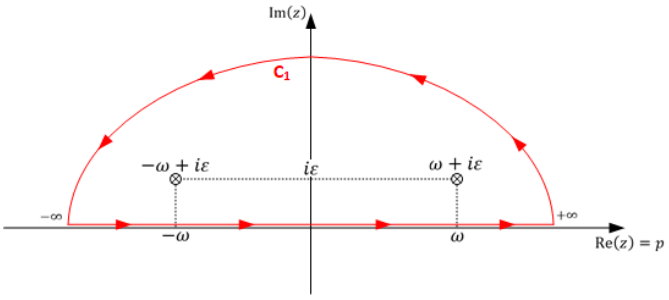


Figure 5: Integration contour C_1 for the advanced propagator

After applying Jordan's lemma and Cauchy's residue theorem we find:

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = \lim_{\varepsilon\to 0^+} i\left[\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}, -\omega+i\varepsilon\right)+\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}, \omega+i\varepsilon\right)\right] \tag{16}$$

The residue at the $-\omega+i\varepsilon$ pole is given by:

$$\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}, -\omega+i\varepsilon\right) = \lim_{z\to-\omega+i\varepsilon}(z-(-\omega+i\varepsilon))\frac{e^{-i(t-t')z}}{(z-i\varepsilon+\omega)(z-i\varepsilon-\omega)}$$

$$\Rightarrow \lim_{\varepsilon\to 0^+} i\,\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}, -\omega+i\varepsilon\right) = \frac{-ie^{i\omega(t-t')}}{2\omega}$$

Similar calculation for the residue at $\omega+i\varepsilon$ yields:

$$\lim_{\varepsilon\to 0^+} i\,\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{(z-i\varepsilon)^2-\omega^2}, \omega+i\varepsilon\right) = \frac{ie^{-i\omega(t-t')}}{2\omega}$$

After inserting these residues in equation (16) we arrive at:

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = \frac{i}{2\omega}\left(e^{-i\omega(t-t')}-e^{i\omega(t-t')}\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = \frac{i}{2\omega}\left(-2i\sin\left(\omega(t-t')\right)\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = \frac{\sin\left(\omega(t-t')\right)}{\omega} \quad (\text{for } t'>t) \tag{17}$$

where in the second line we have again used Euler's formula $e^{-i\alpha}-e^{i\alpha}=-2i\sin\alpha$.
When $t>t'$ we choose contour C_2 in the lower half of the complex plane as shown in Figure 6. Because there are no poles enclosed inside C_2 we can summarily write

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = 0 \quad (\text{for } t>t') \tag{18}$$

The only thing left to do is to combine equations (17) and (18) to obtain:

$$G_-(t'\to t) = \lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{(p-i\varepsilon)^2-\omega^2}\,dp = \begin{cases}0 & \text{for } t>t'\\ \sin\left(\omega(t-t')\right)/\omega & \text{for } t'>t\end{cases} \tag{19}$$

$G_-(t'\to t)$ is the *advanced propagator* from t' to t, also known as the *anti-causal propagator* because it is zero when $t>t'$.

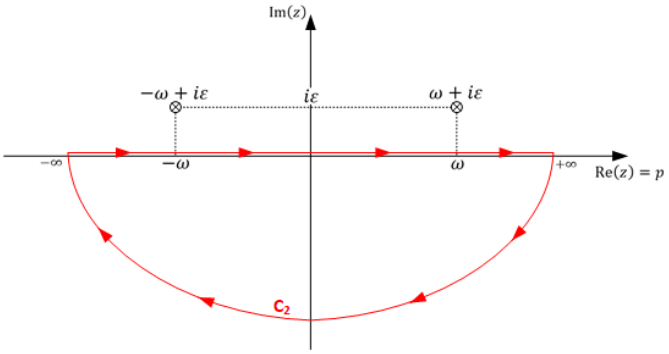


Figure 6: Integration contour C_2 for the advanced propagator

1.3 Feynman propagator

The retarded and advanced propagators have in common that they provide an *all-or-nothing* picture of the system's response: for a given pair of instants t and t', the propagator from t' to t is zero for either $t > t'$ or $t' > t$ (cf. (14) and (19)). By contrast, the *Feynman propagator* constitutes a more balanced approach in which every combination of t and t' yields a non-vanishing result. The Feynman propagator is an essential tool in quantum field theory as it provides the mathematical machinery to incorporate antiparticles into the theory [5]. It is essentially defined by the pole placement shown in Figure 7. The contour integral associated

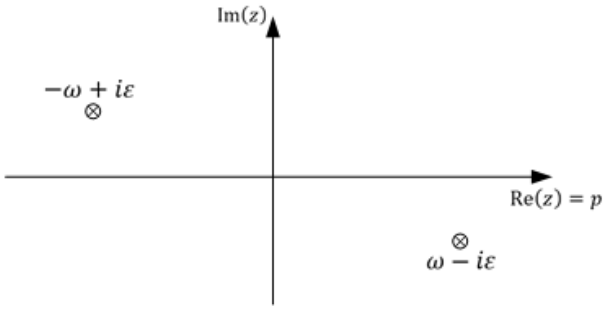


Figure 7: Feynman pole placement

with this pole placement is given by:

$$\lim_{\varepsilon\to 0^+} \frac{1}{2\pi} \oint \frac{e^{-i(t-t')z}}{z^2 - \omega^2 + i\varepsilon}\, dz = \lim_{\varepsilon\to 0^+} \frac{1}{2\pi} \oint \frac{e^{-i(t-t')z}}{\left(z + \omega - \frac{i\varepsilon}{2\omega}\right)\left(z - \omega + \frac{i\varepsilon}{2\omega}\right)}\, dz$$

The denominator can be simplified by rescaling ε by a factor of $1/2\omega$, i.e. $\varepsilon \to \varepsilon/2\omega$, which is allowed because it doesn't impact the limit $\varepsilon \to 0^+$. We can therefore write

$$\lim_{\varepsilon\to 0^+} \frac{1}{2\pi} \oint \frac{e^{-i(t-t')z}}{z^2 - \omega^2 + i\varepsilon}\, dz = \lim_{\varepsilon\to 0^+} \frac{1}{2\pi} \oint \frac{e^{-i(t-t')z}}{(z + \omega - i\varepsilon)(z - \omega + i\varepsilon)}\, dz \tag{20}$$

which clearly shows the poles at $z_1 = -\omega + i\varepsilon$ and $z_2 = \omega - i\varepsilon$, consistent with the placement depicted in Figure 7. The rest of the derivation is very similar to the steps we took for the retarded and advanced propagators. In the $t' > t$ case we choose the C_1 contour which encloses exactly one pole at $-\omega + i\varepsilon$ (cf. Figure 8).

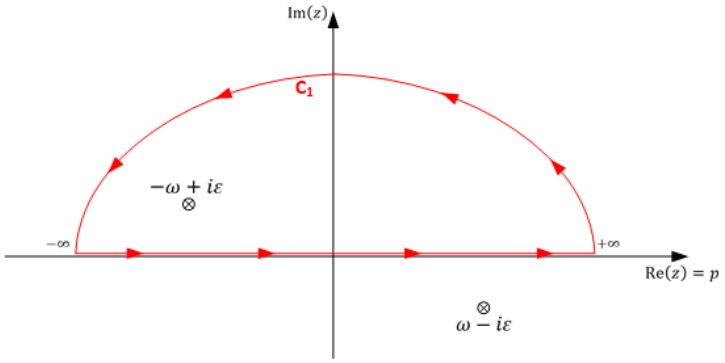


Figure 8: Integration contour C_1 for the Feynman propagator

After applying Jordan's lemma and Cauchy's residue theorem we arrive at:

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\lim_{\varepsilon\to 0^+} i\,\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{z^2-\omega^2+i\varepsilon},-\omega+i\varepsilon\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\lim_{\varepsilon\to 0^+} i\frac{e^{i\omega(t-t')}e^{\varepsilon(t-t')}}{2(i\varepsilon-\omega)}$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\frac{-ie^{i\omega(t-t')}}{2\omega}\quad(\text{for } t'>t) \tag{21}$$

For the $t>t'$ case we choose the C_2 contour in the lower half of the complex plane and pick up the pole at $\omega-i\varepsilon$ (cf. Figure 9):

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\lim_{\varepsilon\to 0^+} -i\,\mathrm{Res}\left(\frac{e^{-i(t-t')z}}{z^2-\omega^2+i\varepsilon},\omega-i\varepsilon\right)$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\lim_{\varepsilon\to 0^+} -i\frac{e^{-i\omega(t-t')}e^{\varepsilon(t-t')}}{2(\omega-i\varepsilon)}$$

$$\lim_{\varepsilon\to 0^+}\frac{1}{2\pi}\int_{-\infty}^{+\infty}\frac{e^{-i(t-t')p}}{p^2-\omega^2+i\varepsilon}\,dp=\frac{-ie^{-i\omega(t-t')}}{2\omega}\quad(\text{for } t>t') \tag{22}$$

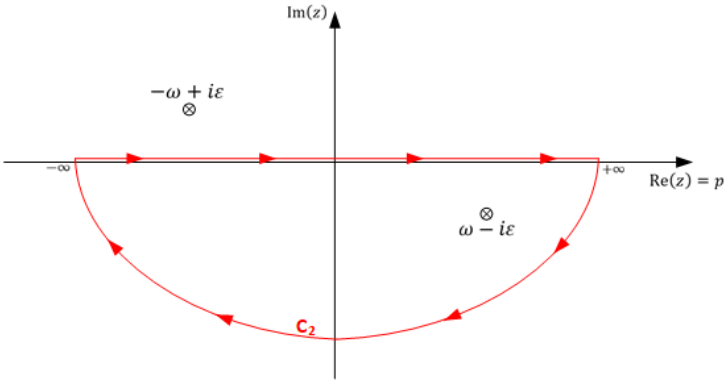


Figure 9: Integration contour C_2 for the Feynman propagator

Combining the results in (21) and (22) yields the expression for the Feynman propagator:

$$G_F(t' \to t) = \lim_{\varepsilon \to 0^+} \frac{1}{2\pi} \int_{-\infty}^{+\infty} \frac{e^{-i(t-t')p}}{p^2 - \omega^2 + i\varepsilon} \, dp = \begin{cases} -ie^{-i\omega(t-t')}/2\omega & \text{for } t > t' \\ -ie^{i\omega(t-t')}/2\omega & \text{for } t' > t \end{cases} \tag{23}$$

Note that the Feynman propagator as defined in (23) is symmetric since:

$$G_F(t \to t') = \begin{cases} -ie^{-i\omega(t'-t)}/2\omega & \text{for } t' > t \\ -ie^{i\omega(t'-t)}/2\omega & \text{for } t > t' \end{cases} = \begin{cases} -ie^{i\omega(t-t')}/2\omega & \text{for } t' > t \\ -ie^{-i\omega(t-t')}/2\omega & \text{for } t > t' \end{cases} = G_F(t' \to t)$$

A natural consequence of this symmetry property is that the Feynman propagator for two instants t and t' only depends on the time difference $t - t'$, which is why we can indiscriminately write $G_F(t' \to t) = G_F(t \to t') = G_F(t - t')$. In relativistic quantum field theory the concept of propagator is generalized to spacetime points and depends on the covariant distance between the two events. For instance $G_F(\boldsymbol{a} - \boldsymbol{b})$ denotes the Feynman propagator between 4-dimensional spacetime points $\boldsymbol{a} = (a^t, a^x, a^y, a^z)$ and $\boldsymbol{b} = (b^t, b^x, b^y, b^z)$.

2 The path integral formulation of quantum mechanics

The path integral formalism is a Lagrangian-based approach to quantum mechanics which departs from the Hamiltonian-based Schrödinger and Heisenberg pictures [6]. Starting from the time-dependent Schrödinger equation

$$i\hbar\frac{d|\psi\rangle}{dt} = \boldsymbol{H}|\psi\rangle \tag{24}$$

we can derive the time-evolution operator $\boldsymbol{U}$ such that:

$$|\psi(t_f)\rangle = \boldsymbol{U}(t_f, t_i)|\psi(t_i)\rangle \quad \text{with} \quad \boldsymbol{U}(t_f, t_i) = e^{-i\boldsymbol{H}(t_f - t_i)/\hbar} \tag{25}$$

provides a solution to (24). For a one-dimensional single particle, the *transition amplitude* that a particle initially located at position x_i at time t_i is later found at position x_f at time t_f is:

$$\langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle = \langle x_f|e^{-i\boldsymbol{H}(t_f - t_i)/\hbar}|x_i\rangle$$

2.1 Time slicing

We now proceed to split the time interval $t_f - t_i$ into N infinitesimal slices of length δt:

$$\begin{gathered}(t_f - t_i) = N\delta t \\ \Rightarrow \langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle = \langle x_f|\underbrace{e^{-i\boldsymbol{H}\delta t/\hbar} \times \cdots \times e^{-i\boldsymbol{H}\delta t/\hbar}}_{N \text{ times}}|x_i\rangle\end{gathered} \tag{26}$$

Let $\boldsymbol{X}$ be the position observable. Because it is a Hermitian operator, its eigenstates form an orthonormal basis which we can use to write a resolution of the identity:

$$\int |x\rangle\langle x|dx = \mathbb{I}$$

We would like to insert this resolution of the identity into (26) after each intermediate infinitesimal time-evolution δt. For the first step we have:

$$\begin{aligned}\langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle &= \langle x_f|e^{-i\boldsymbol{H}\delta t/\hbar} \times \cdots \times e^{-i\boldsymbol{H}\delta t/\hbar}\left(\int |x_1\rangle\langle x_1|dx_1\right)e^{-i\boldsymbol{H}\delta t/\hbar}|x_i\rangle \\ \langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle &= \int dx_1\langle x_f|e^{-i\boldsymbol{H}\delta t/\hbar} \times \cdots \times e^{-i\boldsymbol{H}\delta t/\hbar}|x_1\rangle\langle x_1|e^{-i\boldsymbol{H}\delta t/\hbar}|x_i\rangle\end{aligned}$$

We continue to insert a total of $N - 1$ resolutions of the identity and eventually arrive at the following expression for the transition amplitude:

$$\begin{aligned}\langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle = \int dx_1 \cdots dx_{N-1}\, &\langle x_f|e^{-i\boldsymbol{H}\delta t/\hbar}|x_{N-1}\rangle\langle x_{N-1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_{N-2}\rangle \\ &\times \langle x_{N-2}|e^{-i\boldsymbol{H}\delta t/\hbar} \times \cdots \times e^{-i\boldsymbol{H}\delta t/\hbar}|x_2\rangle \\ &\times \langle x_2|e^{-i\boldsymbol{H}\delta t/\hbar}|x_1\rangle\langle x_1|e^{-i\boldsymbol{H}\delta t/\hbar}|x_i\rangle\end{aligned} \tag{27}$$

Figure 10 provides a visual representation of the time-slicing process that led to (27). It illustrates the fact that the transition amplitude $\langle x_f|\boldsymbol{U}(t_f, t_i)|x_i\rangle$ is calculated as the sum over all possible paths between (x_i, t_i) and (x_f, t_f) in the limit where $N \to \infty$ (i.e. when $\delta t \to 0$).

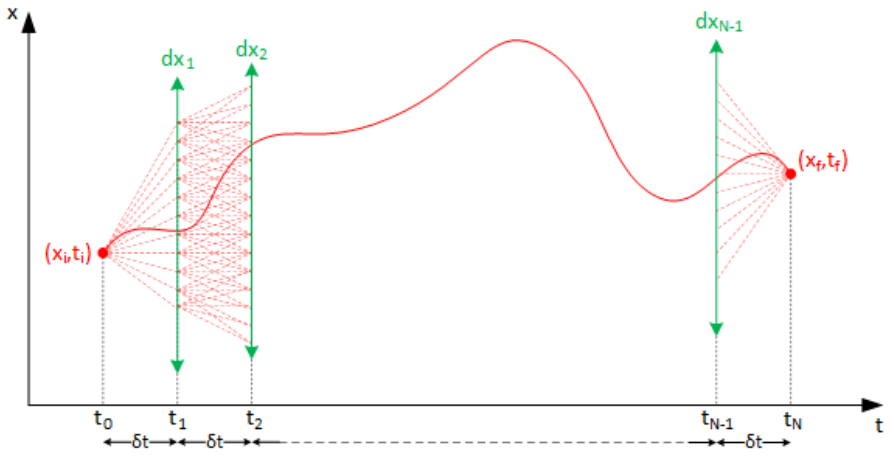


Figure 10: Time slicing of the transition amplitude

2.2 Lagrange enters the stage

In this section we show that each infinitesimal transition amplitude $\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle$ can be rewritten in terms of the Lagrangian

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - V(x) \tag{28}$$

where $V(x)$ represents the particle's potential energy and $\dot{x}$ its velocity. Historically, Dirac was the first to posit that this transition amplitude is *related* to the exponential of the infinitesimal action $\mathcal{L}\delta t$ [7]. Dirac's original intuition greatly influenced Feynman in his derivation of the path integral formulation. We will formally demonstrate this relationship with the generic Hamiltonian observable

$$\boldsymbol{H} = \frac{\boldsymbol{P}^2}{2m} + \boldsymbol{V} \tag{29}$$

where $\boldsymbol{P}$ and $\boldsymbol{V}$ are the momentum and potential energy operators, respectively. Inserting this Hamiltonian into an infinitesimal transition amplitude yields:

$$\begin{aligned}
\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle &= \langle x_{n+1}|e^{-\frac{i}{\hbar}\left(\frac{\boldsymbol{P}^2}{2m}+\boldsymbol{V}\right)\delta t}|x_n\rangle \\
\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle &= \langle x_{n+1}|e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2} \times e^{-\frac{i\delta t}{\hbar}\boldsymbol{V}}|x_n\rangle \\
\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle &= \langle x_{n+1}|e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2} \times e^{-\frac{i\delta t}{\hbar}V(x_n)}|x_n\rangle \\
\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle &= e^{-\frac{i\delta t}{\hbar}V(x_n)}\langle x_{n+1}|e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2}|x_n\rangle
\end{aligned} \tag{30}$$

where in the third line we have used the fact that position eigenstates are also eigenstates of the potential energy observable, i.e. $e^{\boldsymbol{V}}|x_n\rangle = e^{V(x_n)}|x_n\rangle$. To continue we need to decompose $|x_n\rangle$ on the eigenbasis of the momentum observable, which we will rewrite using the position/momentum

representation Fourier transform $\langle p|x\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar}$

$$|x_n\rangle = \int \langle p|x_n\rangle |p\rangle \, dp$$

$$|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_n/\hbar} |p\rangle \, dp$$

$$e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2}|x_n\rangle = e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2} \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_n/\hbar} |p\rangle \, dp$$

$$e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2}|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_n/\hbar} \times e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2} |p\rangle \, dp$$

$$e^{-\frac{i\delta t}{2m\hbar}\boldsymbol{P}^2}|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_n/\hbar} \times e^{-\frac{i\delta t}{2m\hbar}p^2} |p\rangle \, dp$$

Substituting this result directly into (30) gives:

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i\delta t}{\hbar}V(x_n)} \int \langle x_{n+1}|e^{-ipx_n/\hbar} \times e^{-\frac{i\delta t}{2m\hbar}p^2}|p\rangle \, dp$$

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i\delta t}{\hbar}V(x_n)} \int e^{-ipx_n/\hbar} \times e^{-\frac{i\delta t}{2m\hbar}p^2} \langle x_{n+1}|p\rangle \, dp$$

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i\delta t}{\hbar}V(x_n)} \int e^{-ipx_n/\hbar} \times e^{-\frac{i\delta t}{2m\hbar}p^2} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx_{n+1}/\hbar} \, dp$$

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{2\pi\hbar} e^{-\frac{i\delta t}{\hbar}V(x_n)} \int \exp\left(-\frac{i\delta t}{2m\hbar}p^2 + \frac{ip}{\hbar}(x_{n+1} - x_n)\right) dp \tag{31}$$

The integral in (31) is a type of *Gaussian integral* such that [8]:

$$\int \exp\left(-\frac{y^2}{2a} + by\right) dy = \exp\left(\frac{a}{2}b^2\right)\sqrt{2\pi a}$$

Solving this Gaussian integral for $a = m\hbar/i\delta t$ and $b = i(x_{n+1} - x_n)/\hbar$ yields:

$$\int \exp\left(-\frac{i\delta t}{2m\hbar}p^2 + \frac{ip}{\hbar}(x_{n+1} - x_n)\right) dp = \exp\left(\frac{m\hbar}{2i\delta t}\left(\frac{i}{\hbar}(x_{n+1} - x_n)\right)^2\right)\sqrt{\frac{2\pi m\hbar}{i\delta t}}$$

$$\int \exp\left(-\frac{i\delta t}{2m\hbar}p^2 + \frac{ip}{\hbar}(x_{n+1} - x_n)\right) dp = \exp\left(\frac{im}{2\hbar\delta t}(x_{n+1} - x_n)^2\right)\sqrt{\frac{2\pi m\hbar}{i\delta t}}$$

Now we just have to substitute this result into (31) to see how the Lagrangian enters the stage:

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{2\pi\hbar} e^{-\frac{i\delta t}{\hbar}V(x_n)} \exp\left(\frac{im}{2\hbar\delta t}(x_{n+1} - x_n)^2\right)\sqrt{\frac{2\pi m\hbar}{i\delta t}}$$

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \frac{1}{2\pi\hbar}\sqrt{\frac{2\pi m\hbar}{i\delta t}} \exp\left(\frac{im}{2\hbar\delta t}(x_{n+1} - x_n)^2 - \frac{i\delta t}{\hbar}V(x_n)\right)$$

$$\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \exp\left[\frac{i}{\hbar}\left(\frac{1}{2}m\left(\frac{x_{n+1} - x_n}{\delta t}\right)^2 - V(x_n)\right)\delta t\right] \tag{32}$$

In the $\delta t \to 0$ limit, the term $(x_{n+1} - x_n)/\delta t$ is easily identified as the instantaneous velocity $\dot{x}$ and thus we have:

$$\lim_{\delta t \to 0}\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \exp\left[\frac{i}{\hbar}\left(\frac{1}{2}m\dot{x}^2 - V(x_n)\right)\delta t\right]$$

$$\Leftrightarrow \lim_{\delta t\to 0}\langle x_{n+1}|e^{-i\boldsymbol{H}\delta t/\hbar}|x_n\rangle = \sqrt{\frac{m}{i\delta t 2\pi\hbar}}\exp\left(\frac{i}{\hbar}\mathcal{L}\delta t\right) \tag{33}$$

where in the second line we have used the Lagrangian definition in (28). The infinitesimal transition amplitude is not just *related* to the exponential of the infinitesimal action $\mathcal{L}\delta t$ as Dirac predicted. It is actually proportional to it and the proportionality factor only depends on the particle's mass, and not on its trajectory or any other dynamic variable.

2.3 The path integral equation

To complete the path integral formulation we must insert the infinitesimal transition amplitude (33) into the expression (27) for the full transition amplitude:

$$\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle = \lim_{\substack{\delta t\to 0\\ N\to\infty}}\int dx_1\cdots dx_{N-1}\,\langle x_f|e^{-i\boldsymbol{H}\delta t/\hbar}|x_{N-1}\rangle\times\cdots\times\langle x_1|e^{-i\boldsymbol{H}\delta t/\hbar}|x_i\rangle$$

$$\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle = \lim_{\substack{\delta t\to 0\\ N\to\infty}}\int dx_1\cdots dx_{N-1}\underbrace{\sqrt{\frac{m}{i\delta t 2\pi\hbar}}\exp\left(\frac{i}{\hbar}\mathcal{L}\delta t\right)\times\cdots\times\sqrt{\frac{m}{i\delta t 2\pi\hbar}}\exp\left(\frac{i}{\hbar}\mathcal{L}\delta t\right)}_{\text{from }(x_i,t_i)\text{ to }(x_f,t_f)}$$

$$\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle = \lim_{\substack{\delta t\to 0\\ N\to\infty}}\left(\frac{m}{i\delta t 2\pi\hbar}\right)^{N/2}\int dx_1\cdots dx_{N-1}\exp\left(\frac{i}{\hbar}\sum_{t_i}^{t_f}\mathcal{L}\delta t\right)$$

In the $\delta t\to 0$ limit the discrete sum $\sum_{t_i}^{t_f}\mathcal{L}\delta t$ becomes a continuous integral $\int_{t_i}^{t_f}\mathcal{L}dt$, which is precisely the action $S[x(t)]$ between (x_i,t_i) and (x_f,t_f). In a more succinct form we can write:

$$\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle = K(m,\delta t)\int_{x(t_i)=x_i}^{x(t_f)=x_f}\mathcal{D}[x(t)]\exp\left(\frac{i}{\hbar}S[x(t)]\right) \tag{34}$$

where the symbol $\int\mathcal{D}[x(t)]\simeq\int dx_1\cdots dx_{N-1}$ signifies that we integrate over all possible paths between (x_i,t_i) and (x_f,t_f), and we have introduced the constant

$$K(m,\delta t) = \lim_{\substack{\delta t\to 0\\ N\to\infty}}\left(\frac{m}{i\delta t 2\pi\hbar}\right)^{N/2}$$

The divergence of this constant in the limit $\delta t\to 0$ and $N\to\infty$ is discussed in section 2.6.

2.4 Path integral and propagators

$\boldsymbol{U}(t_f,t_i)$ is a time-evolution operator, and $\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle$ is the (x_f,x_i) element of its matrix representation in the basis of position eigenstates. We will often use the notation $U(x_f,t_f,x_i,t_i)$ to represent the time-evolution matrix element $\langle x_f|\boldsymbol{U}(t_f,t_i)|x_i\rangle$. Suppose we know the state of a particle at time t_0 but would like to evaluate its wave function at a later time $t>t_0$. From the definition (25) of the time-evolution operator we have:

$$|\psi(t)\rangle = \boldsymbol{U}(t,t_0)|\psi(t_0)\rangle$$

which after inserting the resolution of the identity $\int|x'\rangle\langle x'|dx' = \mathbb{I}$ becomes:

$$|\psi(t)\rangle = \int\boldsymbol{U}(t,t_0)|x'\rangle\langle x'|\psi(t_0)\rangle\,dx'$$

To find the wave function in the position representation, we simply have to project this equation onto the position eigenbra $\langle x|$

$$\langle x|\psi(t)\rangle = \langle x| \int \boldsymbol{U}(t,t_0)|x'\rangle\langle x'|\psi(t_0)\rangle\, dx'$$

$$\psi(x,t) = \int \langle x|\boldsymbol{U}(t,t_0)|x'\rangle\langle x'|\psi(t_0)\rangle\, dx'$$

$$\psi(x,t) = \int U(x,t,x',t_0)\psi(x',t_0)\, dx' \tag{35}$$

The last equation resembles the Green function's definition given in (5), which tells us that $U(x,t,x',t_0)$ can be interpreted as the propagator from (x',t_0) to (x,t). Accordingly, the integrand $U(x,t,x',t_0)\psi(x',t_0)$ represents the wave function propagated from (x',t_0) to (x,t). The upshot of the equality in (35) is that we can build the wave function $\psi(x)$ at time t as the sum over every possible trajectory between t_0 and t. Specifically, the integral $\int dx'$ sums over all possible starting position of the particle, while $U(x,t,x',t_0)$ sums over all possible paths between (x',t_0) and (x,t).

2.5 Equivalence with the Schrödinger equation

Consider the infinitesimal form of the path integral, which we know from (32) to be given by:

$$U(x,\delta t,x_0,0) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \exp\left[\frac{i}{\hbar}\left(\frac{1}{2}m\left(\frac{x-x_0}{\delta t}\right)^2 - V(x_0)\right)\delta t\right]$$

In the limit $\delta t \to 0$ we have $x \to x_0$, and therefore we can reasonably assume that $V(x) \approx V(x_0)$ and write:

$$U(x,\delta t,x_0,0) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \exp\left[\frac{i}{\hbar}\left(\frac{1}{2}m\left(\frac{x-x_0}{\delta t}\right)^2 - V(x)\right)\delta t\right]$$

$$U(x,\delta t,x_0,0) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \exp\left[\frac{i}{\hbar}\left(\frac{m}{2\delta t}(x-x_0)^2 - V(x)\delta t\right)\right]$$

We can substitute this result into equation (35) to get:

$$\psi(x,\delta t) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \int \exp\left[\frac{i}{\hbar}\left(\frac{m}{2\delta t}(x-x_0)^2 - V(x)\delta t\right)\right]\psi(x_0,0)\, dx_0$$

$$\psi(x,\delta t) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}} \int \exp\left[\frac{i}{\hbar}\left(\frac{m}{2\delta t}\eta^2 - V(x)\delta t\right)\right]\psi(x+\eta,0)\, d\eta \tag{36}$$

where in the last equation we changed the integration variable to $\eta = x_0 - x$. In the limit $\delta t \to 0$ we can write the first order expansion $\exp\left(-\frac{i}{\hbar}V(x)\delta t\right) = \left(1 - \frac{i}{\hbar}V(x)\delta t\right)$ and therefore we have:

$$\exp\left[\frac{i}{\hbar}\left(\frac{m}{2\delta t}\eta^2 - V(x)\delta t\right)\right] = \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\left(1 - \frac{i}{\hbar}V(x)\delta t\right)$$

$$\Rightarrow \psi(x,\delta t) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}}\left(1 - \frac{i}{\hbar}V(x)\delta t\right)\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\psi(x+\eta,0)\, d\eta \tag{37}$$

Again, in the limit $\delta t \to 0$ we have $x \to x_0 \Rightarrow \eta = x_0 - x \to 0$ and we can write the second order expansion

$$\psi(x+\eta,0) = \psi(x,0) + \eta\frac{\partial\psi(x,0)}{\partial x} + \frac{\eta^2}{2}\frac{\partial^2\psi(x,0)}{\partial x^2}$$

$$\Rightarrow \psi(x,\delta t) = \sqrt{\frac{m}{i\delta t 2\pi\hbar}}\left(1 - \frac{i}{\hbar}V(x)\delta t\right)\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\left(\psi(x,0) + \eta\frac{\partial\psi(x,0)}{\partial x} + \frac{\eta^2}{2}\frac{\partial^2\psi(x,0)}{\partial x^2}\right)d\eta$$

$$\psi(x,\delta t) = \sqrt{\frac{m}{2\pi\hbar i\delta t}}\left(1-\frac{i\delta t}{\hbar}V(x)\right)\left[\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\psi(x,0)\,d\eta \right.$$
$$+\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\eta\frac{\partial\psi(x,0)}{\partial x}\,d\eta \qquad (38)$$
$$\left.+\frac{1}{2}\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\eta^2\frac{\partial^2\psi(x,0)}{\partial x^2}\,d\eta\right]$$

We tackle each integral in turn.

Zeroth-order integral

$$A_0 = \int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\psi(x,0)\,d\eta$$

Since $\psi(x,0)$ does not depend on the integration variable η, we can write

$$A_0 = \psi(x,0)\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)d\eta$$

This is yet another Gaussian integral of the form [8]:

$$\int \exp\left(-\frac{y^2}{2a}\right)dy = \sqrt{2\pi a}$$

which for $a = -\hbar\delta t/im$ yields:

$$A_0 = \psi(x,0)\sqrt{\frac{-2\pi\hbar\delta t}{im}} = \psi(x,0)\sqrt{\frac{2\pi\hbar i\delta t}{m}} \qquad (39)$$

First-order integral

$$A_1 = \int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\eta\frac{\partial\psi(x,0)}{\partial x}\,d\eta = \frac{\partial\psi(x,0)}{\partial x}\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\eta\,d\eta$$

The integrand is clearly an odd function of η: it is the product of the odd function η and the even function $\exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)$. Since for any odd function $f(\eta)$ we have:

$$f(-\eta) = -f(\eta) \Rightarrow \int_{-\infty}^{+\infty} f(\eta)d\eta = 0$$

we conclude that $A_1 = 0$.

Second-order integral

$$A_2 = \frac{1}{2}\frac{\partial^2\psi(x,0)}{\partial x^2}\int \exp\left(\frac{im\eta^2}{2\hbar\delta t}\right)\eta^2\,d\eta$$

This is a variation of the Gaussian integral of the form which we can evaluate by differentiating the standard Gaussian integral $\int \exp\left(-\frac{y^2}{2a}\right)dy = \sqrt{2\pi a}$ with respect to a:

$$\frac{\partial}{\partial a}\left(\int \exp\left(-\frac{y^2}{2a}\right)dy\right) = \frac{\partial}{\partial a}\left(\sqrt{2\pi a}\right)$$

$$\int \frac{y^2}{2a^2}\exp\left(-\frac{y^2}{2a}\right)dy = \frac{1}{2}\sqrt{\frac{2\pi}{a}}$$

$$\int \exp\left(-\frac{y^2}{2a}\right)y^2 dy = \frac{2a^2}{2}\sqrt{\frac{2\pi}{a}} = a\sqrt{2\pi a}$$

With this result it becomes straightforward to evaluate the second-order integral A_2. We simply need to replace a with $-\hbar\delta t/im$ to find:

$$A_2 = \frac{1}{2}\frac{\partial^2\psi(x,0)}{\partial x^2}\left(\frac{-\hbar\delta t}{im}\right)\sqrt{\frac{-2\pi\hbar\delta t}{im}} = \frac{\partial^2\psi(x,0)}{\partial x^2}\frac{i\hbar\delta t}{2m}\sqrt{\frac{2\pi\hbar i\delta t}{m}} \tag{40}$$

Inserting these expressions for A_0, A_1, and A_2 into equation (38) yields:

$$\psi(x,\delta t) = \sqrt{\frac{m}{2\pi\hbar i\delta t}}\left(1-\frac{i\delta t}{\hbar}V(x)\right)\left[\psi(x,0)\sqrt{\frac{2\pi\hbar i\delta t}{m}} + \frac{\partial^2\psi(x,0)}{\partial x^2}\frac{i\hbar\delta t}{2m}\sqrt{\frac{2\pi\hbar i\delta t}{m}}\right]$$

$$\psi(x,\delta t) = \sqrt{\frac{m}{2\pi\hbar i\delta t}}\times\sqrt{\frac{2\pi\hbar i\delta t}{m}}\left(1-\frac{i\delta t}{\hbar}V(x)\right)\left[\psi(x,0) + \frac{\partial^2\psi(x,0)}{\partial x^2}\frac{i\hbar\delta t}{2m}\right]$$

$$\psi(x,\delta t) = \left(1-\frac{i\delta t}{\hbar}V(x)\right)\left[\psi(x,0) + \frac{\partial^2\psi(x,0)}{\partial x^2}\frac{i\hbar\delta t}{2m}\right]$$

The annoying constant $\sqrt{m/2\pi\hbar i\delta t}$, whose divergence in the limit $\delta t \to 0$ should have filled us with dread when we derived equation (34), has thankfully been canceled out by an inverse factor coming from A_0 and A_2. This *natural* cancellation tells us that we should not be overly concerned about this diverging constant. If we stick to first-order terms in δt we are left with:

$$\psi(x,\delta t) = \psi(x,0) + \frac{i\hbar\delta t}{2m}\frac{\partial^2\psi(x,0)}{\partial x^2} - \frac{i\delta t}{\hbar}V(x)\psi(x,0)$$

$$i\hbar\frac{\psi(x,\delta t)-\psi(x,0)}{\delta t} = \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi(x,0)$$

which the limit $\delta t \to 0$ matches exactly the time-dependent Schrödinger equation:

$$i\hbar\frac{\partial\psi}{\partial t} = \left(\frac{\boldsymbol{P}^2}{2m} + \boldsymbol{V}\right)\psi$$

2.6 Normalization

As we saw in the previous section where we derived the Schrödinger equation from the path integral, the constant

$$K(m,\delta t) = \lim_{\substack{\delta t\to 0\\ N\to\infty}}\left(\frac{m}{i\delta t 2\pi\hbar}\right)^{N/2}$$

sitting in front of the path integral is factored out when we calculate physical quantities. In practice, generic propagators like $U(x_f,t_f,x_i,t_i)$ are normalized with respect to the free particle propagator denoted $U_0(x_f,t_f,x_i,t_i)$. The idea is to multiply $U(x_f,t_f,x_i,t_i)$ by the unit fraction

$$\frac{U_0(x_f,t_f,x_i,t_i)}{U_0(x_f,t_f,x_i,t_i)} = 1$$

Let $\mathcal{U}$ be the *normalized* propagator defined as:

$$\mathcal{U}(x_f,t_f,x_i,t_i) = \frac{U_0(x_f,t_f,x_i,t_i)}{U_0(x_f,t_f,x_i,t_i)}U(x_f,t_f,x_i,t_i)$$

$$\mathcal{U}(x_f,t_f,x_i,t_i) = U_0(x_f,t_f,x_i,t_i)\frac{\lim_{\substack{\delta t\to 0\\ N\to\infty}}\left(\frac{m}{i\delta t 2\pi\hbar}\right)^{N/2}\int\mathcal{D}[x(t)]\exp\left(\frac{i}{\hbar}S[x(t)]\right)}{\lim_{\substack{\delta t\to 0\\ N\to\infty}}\left(\frac{m}{i\delta t 2\pi\hbar}\right)^{N/2}\int\mathcal{D}[x(t)]\exp\left(\frac{i}{\hbar}S_0[x(t)]\right)}$$

$$\mathcal{U}(x_f, t_f, x_i, t_i) = U_0(x_f, t_f, x_i, t_i) \frac{\int \mathcal{D}[x(t)] \exp\left(\frac{i}{\hbar} S[x(t)]\right)}{\int \mathcal{D}[x(t)] \exp\left(\frac{i}{\hbar} S_0[x(t)]\right)} \tag{41}$$

where $S_0[x(t)] = \int \frac{1}{2} m\dot{x}^2 \, dt$ is the free particle action. The free propagator $U_0(x_f, t_f, x_i, t_i)$ can be explicitly calculated as a Gaussian integral:

$$U_0(x_f, t_f, x_i, t_i) = \langle x_f | \boldsymbol{U}_0(t_f, t_i) | x_i \rangle = \langle x_f | e^{-i\boldsymbol{H}_0(t_f - t_i)/\hbar} | x_i \rangle$$

$$U_0(x_f, t_f, x_i, t_i) = \langle x_f | e^{-i\boldsymbol{P}^2(t_f - t_i)/(2m\hbar)} | x_i \rangle$$

We can expand the position eigenstate $|x_i\rangle$ in the momentum eigenbasis (cf. section 2.2)

$$|x_i\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_i/\hbar} |p\rangle \, dp$$

and insert it into our previous expression for $U_0(x_f, t_f, x_i, t_i)$:

$$U_0(x_f, t_f, x_i, t_i) = \langle x_f | e^{-i\boldsymbol{P}^2(t_f - t_i)/(2m\hbar)} \int e^{-ipx_i/\hbar} |p\rangle \, dp$$

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_i/\hbar} \langle x_f | e^{-i\boldsymbol{P}^2(t_f - t_i)/(2m\hbar)} |p\rangle \, dp$$

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_i/\hbar} \langle x_f | e^{-ip^2(t_f - t_i)/(2m\hbar)} |p\rangle \, dp$$

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_i/\hbar} \times e^{-ip^2(t_f - t_i)/(2m\hbar)} \langle x_f | p\rangle \, dp$$

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx_i/\hbar} \times e^{-ip^2(t_f - t_i)/(2m\hbar)} \frac{1}{\sqrt{2\pi\hbar}} e^{ipx_f/\hbar} \, dp$$

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{2\pi\hbar} \int \exp\left(-\frac{i}{\hbar} \frac{p^2}{2m} (t_f - t_i) + \frac{ip}{\hbar} (x_f - x_i) \right) dp$$

We are again faced with a Gaussian integral of the form

$$\int \exp\left(-\frac{p^2}{2a} + bp \right) dp = \sqrt{2\pi a} \, \exp\left(\frac{a}{2} b^2 \right)$$

which for $a = m\hbar / i(t_f - t_i)$ and $b = i(x_f - x_i)/\hbar$ yields:

$$U_0(x_f, t_f, x_i, t_i) = \frac{1}{2\pi\hbar} \exp\left(\frac{m\hbar}{2i(t_f - t_i)} \left(\frac{i(x_f - x_i)}{\hbar} \right)^2 \right) \sqrt{\frac{2\pi m\hbar}{i(t_f - t_i)}}$$

$$U_0(x_f, t_f, x_i, t_i) = \sqrt{\frac{m}{i(t_f - t_i) 2\pi\hbar}} \exp\left(\frac{im(x_f - x_i)^2}{2\hbar(t_f - t_i)} \right)$$

Substituting this expression for $U_0(x_f, t_f, x_i, t_i)$ into equation (41) finally gives us:

$$U(x_f, t_f, x_i, t_i) = \sqrt{\frac{m}{i(t_f - t_i) 2\pi\hbar}} \exp\left(\frac{im(x_f - x_i)^2}{2\hbar(t_f - t_i)} \right) \frac{\int \mathcal{D}[x(t)] e^{iS[x(t)]/\hbar}}{\int \mathcal{D}[x(t)] e^{iS_0[x(t)]/\hbar}} \tag{42}$$

The diverging constant $K(m, \delta t)$ has been factored out, as promised.

3 Path integrals in quantum field theory

3.1 Time slicing of field configurations

The generalization of path integrals to fields is relatively straightforward. Instead of time-slicing a particle movement over all possible paths in space, the field is time-sliced over all possible field configurations. The transition amplitude for the field to go from an initial configuration $|\psi_i\rangle$ at time t_i to a final configuration $|\psi_f\rangle$ at time t_f is given by:

$$\langle\psi_f|\boldsymbol{U}(t_f,t_i)|\psi_i\rangle = \int_{\psi(\vec{x},t_i)=\psi_i(\vec{x})}^{\psi(\vec{x},t_f)=\psi_f(\vec{x})} \mathcal{D}[\psi(\vec{x},t)]\, e^{iS[\psi(\vec{x},t)]/\hbar} \tag{43}$$

The action for the field is defined as

$$S[\psi(\vec{x},t)] = \int_{t_i}^{t_f} dt \int \mathcal{L}(\psi,\partial_\mu\psi)\, d^3x \tag{44}$$

where $\partial_\mu\psi$ represents the partial derivatives of the field with respect to spacetime coordinates $x^\mu = (t,x,y,z)$. Said differently, $\partial_\mu = (\partial_t, \vec{\nabla})$ is the four-gradient operator in Minkowski spacetime. Writing the time-slicing explicitly equation (43) becomes:

$$\begin{aligned}\langle\psi_f|\boldsymbol{U}(t_f,t_i)|\psi_i\rangle = \int d\psi_1(\vec{x})\cdots d\psi_{N-1}(\vec{x})\, &\langle\psi_f|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_{N-1}\rangle \\ &\times \langle\psi_{N-1}|e^{-i\boldsymbol{H}\delta t/\hbar}\times\cdots\times e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_2\rangle \\ &\times \langle\psi_2|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_1\rangle\langle\psi_1|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_i\rangle\end{aligned}$$

The integration variable $d\psi_n(\vec{x})$ *sums* over all possible field configurations at time-slice n. A field configuration corresponds to a particular field value at every point in space, which is why it is written $d\psi_n(\vec{x})$.

3.2 From path integrals to Feynman propagators

Consider the transition amplitude $\langle\phi|\boldsymbol{U}(t_f,t_i)|\phi\rangle$ for a scalar field, where the initial and final field configurations are the same:

$$\langle\phi|\boldsymbol{U}(t_f,t_i)|\phi\rangle = \int_{|\psi(t_i)\rangle=|\phi\rangle}^{|\psi(t_f)\rangle=|\phi\rangle} \mathcal{D}[\psi(\vec{x},t)]\, e^{iS[\psi(\vec{x},t)]/\hbar}$$

$$\begin{aligned}\langle\phi|\boldsymbol{U}(t_f,t_i)|\phi\rangle = \int d\psi_1(\vec{x})\cdots d\psi_{N-1}(\vec{x})\, &\langle\phi|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_{N-1}\rangle \\ &\times \langle\psi_{N-1}|e^{-i\boldsymbol{H}\delta t/\hbar}\cdots e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_1\rangle \\ &\times \langle\psi_1|e^{-i\boldsymbol{H}\delta t/\hbar}|\phi\rangle\end{aligned}$$

Assuming $t_f > t_k > t_n > t_i$, we would like to calculate the quantity $\mathcal{P}$ defined by:

$$\mathcal{P} = \int_{|\psi(t_i)\rangle=|\phi\rangle}^{|\psi(t_f)\rangle=|\phi\rangle} \mathcal{D}[\psi(\vec{x},t)]\, e^{iS[\psi(\vec{x},t)]/\hbar}\, \psi(\vec{x}_n,t_n)\psi(\vec{x}_k,t_k) \tag{45}$$

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x})\cdots d\psi_{N-1}(\vec{x})\, &\langle\phi|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_{N-1}\rangle \\ &\times \langle\psi_{N-1}|e^{-i\boldsymbol{H}\delta t/\hbar}\cdots e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_1\rangle \\ &\times \langle\psi_1|e^{-i\boldsymbol{H}\delta t/\hbar}|\phi\rangle\psi(\vec{x}_n,t_n)\psi(\vec{x}_k,t_k)\end{aligned}$$

where $\psi(\vec{x}_n, t_n)$ and $\psi(\vec{x}_k, t_k)$ represent the field scalar values at spacetime points $(t_n, \vec{x}_n)$ and $(t_k, \vec{x}_k)$ respectively. Since we are considering the case where $t_f > t_k > t_n > t_i$, we can explicitly write the time-slices n and k as follows:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x}) \cdots d\psi_n(\vec{x}) \cdots d\psi_k(\vec{x}) \cdots d\psi_{N-1}(\vec{x}) \times \langle \phi | e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_{N-1} \rangle \\ \times \langle \psi_{N-1} | e^{-i\boldsymbol{H}\delta t/\hbar} \cdots e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_k \rangle \\ \times \langle \psi_k | e^{-i\boldsymbol{H}\delta t/\hbar} \cdots e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_n \rangle \\ \times \langle \psi_n | e^{-i\boldsymbol{H}\delta t/\hbar} \cdots e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_1 \rangle \\ \times \langle \psi_1 | e^{-i\boldsymbol{H}\delta t/\hbar} | \phi \rangle \, \psi(\vec{x}_n, t_n) \psi(\vec{x}_k, t_k)\end{aligned}$$

Notice how, to be consistent with the $t_f > t_k > t_n > t_i$ assumption, the k time-slice has been put *later* than the n time-slice: i.e. k is closer to the last time-slice $N-1$ than n is. The various $e^{-i\boldsymbol{H}\delta t/\hbar} \cdots e^{-i\boldsymbol{H}\delta t/\hbar}$ can be simplified by counting the number of infinitesimal steps of δt between the two surrounding time-slices:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x}) \cdots d\psi_n(\vec{x}) \cdots d\psi_k(\vec{x}) \cdots d\psi_{N-1}(\vec{x}) \times \langle \phi | e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_{N-1} \rangle \\ \times \langle \psi_{N-1} | e^{-i\boldsymbol{H}(t_{N-1} - t_k)/\hbar} | \psi_k \rangle \\ \times \langle \psi_k | e^{-i\boldsymbol{H}(t_k - t_n)/\hbar} | \psi_n \rangle \\ \times \langle \psi_n | e^{-i\boldsymbol{H}(t_n - t_1)/\hbar} | \psi_1 \rangle \\ \times \langle \psi_1 | e^{-i\boldsymbol{H}\delta t/\hbar} | \phi \rangle \, \psi(\vec{x}_n, t_n) \psi(\vec{x}_k, t_k)\end{aligned}$$

Because $\psi(\vec{x}_n, t_n)$ and $\psi(\vec{x}_k, t_k)$ are scalars, we are free to move them anywhere inside the product of infinitesimal transition amplitudes without worrying about commutation relations. We strategically choose to move $\psi(\vec{x}_n, t_n)$ to the left of $|\psi_n\rangle$, and $\psi(\vec{x}_k, t_k)$ to the right of $\langle \psi_k |$:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x}) \cdots d\psi_n(\vec{x}) \cdots d\psi_k(\vec{x}) \cdots d\psi_{N-1}(\vec{x}) \times \langle \phi | e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_{N-1} \rangle \\ \times \langle \psi_{N-1} | e^{-i\boldsymbol{H}(t_{N-1} - t_k)/\hbar} | \psi_k \rangle \\ \times \langle \psi_k | \psi(\vec{x}_k, t_k) e^{-i\boldsymbol{H}(t_k - t_n)/\hbar} \psi(\vec{x}_n, t_n) | \psi_n \rangle \\ \times \langle \psi_n | e^{-i\boldsymbol{H}(t_n - t_1)/\hbar} | \psi_1 \rangle \\ \times \langle \psi_1 | e^{-i\boldsymbol{H}\delta t/\hbar} | \phi \rangle\end{aligned}$$

In the Schrödinger picture, we define $\hat{\psi}(\vec{x}_k)$ to be the *field operator* whose eigenvalue associated to the field configuration $|\psi_k\rangle$ is the field value at spacetime point $(t_k, \vec{x}_k)$:

$$\hat{\psi}(\vec{x}_k) |\psi_k\rangle = \psi(\vec{x}_k, t_k) |\psi_k\rangle \Leftrightarrow \langle \psi_k | \hat{\psi}(\vec{x}_k) = \langle \psi_k | \psi(\vec{x}_k, t_k)$$

We then substitute $\langle \psi_k | \psi(\vec{x}_k, t_k)$ with $\langle \psi_k | \hat{\psi}(\vec{x}_k)$, and $\psi(\vec{x}_n, t_n) | \psi_n \rangle$ with $\hat{\psi}(\vec{x}_n) | \psi_n \rangle$ in our last expression for $\mathcal{P}$ to get:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x}) \cdots d\psi_n(\vec{x}) \cdots d\psi_k(\vec{x}) \cdots d\psi_{N-1}(\vec{x}) \times \langle \phi | e^{-i\boldsymbol{H}\delta t/\hbar} | \psi_{N-1} \rangle \\ \times \langle \psi_{N-1} | e^{-i\boldsymbol{H}(t_{N-1} - t_k)/\hbar} | \psi_k \rangle \\ \times \langle \psi_k | \hat{\psi}(\vec{x}_k) e^{-i\boldsymbol{H}(t_k - t_n)/\hbar} \hat{\psi}(\vec{x}_n) | \psi_n \rangle \\ \times \langle \psi_n | e^{-i\boldsymbol{H}(t_n - t_1)/\hbar} | \psi_1 \rangle \\ \times \langle \psi_1 | e^{-i\boldsymbol{H}\delta t/\hbar} | \phi \rangle\end{aligned}$$

Buried inside this integral are $N-1$ resolutions of the identity of the form

$$\int d\psi_p(\vec{x})|\psi_p\rangle\langle\psi_p| = \mathbb{I} \quad p \in [1..(N-1)]$$

which can be used to get rid of every integral and reduce the expression for $\mathcal{P}$ to:

$$\mathcal{P} = \langle\phi|e^{-i\boldsymbol{H}(t_f-t_k)/\hbar}\,\hat{\psi}(\vec{x}_k)\,e^{-i\boldsymbol{H}(t_k-t_n)/\hbar}\,\hat{\psi}(\vec{x}_n)\,e^{-i\boldsymbol{H}(t_n-t_i)/\hbar}|\phi\rangle$$

$$\Leftrightarrow \mathcal{P} = \langle\phi|e^{-it_f\boldsymbol{H}/\hbar}\,e^{it_k\boldsymbol{H}/\hbar}\,\hat{\psi}(\vec{x}_k)\,e^{-it_k\boldsymbol{H}/\hbar}\,e^{it_n\boldsymbol{H}/\hbar}\,\hat{\psi}(\vec{x}_n)\,e^{-it_n\boldsymbol{H}/\hbar}\,e^{it_i\boldsymbol{H}/\hbar}|\phi\rangle$$

We recognize Heisenberg picture field operators in the last equation, namely:

$$\hat{\psi}(\boldsymbol{x}_k)_H = e^{it_k\boldsymbol{H}/\hbar}\,\hat{\psi}(\vec{x}_k)\,e^{-it_k\boldsymbol{H}/\hbar}$$

$$\hat{\psi}(\boldsymbol{x}_n)_H = e^{it_n\boldsymbol{H}/\hbar}\,\hat{\psi}(\vec{x}_n)\,e^{-it_n\boldsymbol{H}/\hbar}$$

where we have defined the spacetime points $\boldsymbol{x}_k = (t_k, \vec{x}_k)$ and $\boldsymbol{x}_n = (t_n, \vec{x}_n)$. Our final expression for $\mathcal{P}$ which we defined in (45) is therefore[1]:

$$\int \mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) = \langle\phi|e^{-it_f\boldsymbol{H}/\hbar}\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)e^{it_i\boldsymbol{H}/\hbar}|\phi\rangle \quad (t_k > t_n) \tag{46}$$

where we dropped the $_H$ subscript for brevity, and the path integral is understood to be taken between field configurations $|\psi(t_i)\rangle = |\phi\rangle$ and $|\psi(t_f)\rangle = |\phi\rangle$

We can go through the same exercise with the reverse assumption $t_n > t_k$. Consider the same quantity $\mathcal{P}$ defined in (45):

$$\mathcal{P} = \int_{|\psi(t_i)\rangle=|\phi\rangle}^{|\psi(t_f)\rangle=|\phi\rangle} \mathcal{D}[\psi(\vec{x},t)]\,e^{iS[\psi(\vec{x},t)]/\hbar}\,\psi(\vec{x}_n,t_n)\psi(\vec{x}_k,t_k)$$

only this time because $t_f > t_n > t_k > t_i$ the order in which time-slices k and n appear in the product is reversed:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x})\cdots d\psi_n(\vec{x})\cdots d\psi_k(\vec{x})\cdots d\psi_{N-1}(\vec{x}) \times\ & \langle\phi|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_{N-1}\rangle \\ & \times \langle\psi_{N-1}|e^{-i\boldsymbol{H}(t_{N-1}-t_n)/\hbar}|\psi_n\rangle \\ & \times \langle\psi_n|e^{-i\boldsymbol{H}(t_n-t_k)/\hbar}|\psi_k\rangle \\ & \times \langle\psi_k|e^{-i\boldsymbol{H}(t_k-t_1)/\hbar}|\psi_1\rangle \\ & \times \langle\psi_1|e^{-i\boldsymbol{H}\delta t/\hbar}|\phi\rangle\ \psi(\vec{x}_n,t_n)\psi(\vec{x}_k,t_k)\end{aligned}$$

We can still move the scalar field values $\psi(\vec{x}_n,t_n)$ and $\psi(\vec{x}_k,t_k)$ next to $\langle\psi_n|$ and $|\psi_k\rangle$, respectively, and replace them with the Schrödinger picture field operators $\hat{\psi}(\vec{x}_n)$ and $\hat{\psi}(\vec{x}_k)$:

$$\begin{aligned}\mathcal{P} = \int d\psi_1(\vec{x})\cdots d\psi_n(\vec{x})\cdots d\psi_k(\vec{x})\cdots d\psi_{N-1}(\vec{x}) \times\ & \langle\phi|e^{-i\boldsymbol{H}\delta t/\hbar}|\psi_{N-1}\rangle \\ & \times \langle\psi_{N-1}|e^{-i\boldsymbol{H}(t_{N-1}-t_n)/\hbar}|\psi_n\rangle \\ & \times \langle\psi_n|\hat{\psi}(\vec{x}_n)e^{-i\boldsymbol{H}(t_n-t_k)/\hbar}\hat{\psi}(\vec{x}_k)|\psi_k\rangle \\ & \times \langle\psi_k|e^{-i\boldsymbol{H}(t_k-t_1)/\hbar}|\psi_1\rangle \\ & \times \langle\psi_1|e^{-i\boldsymbol{H}\delta t/\hbar}|\phi\rangle\end{aligned}$$

[1] 3-vectors are written with an explicit arrow $\vec{x}$, and 4-vectors with lower case bold letters $\boldsymbol{x}$

After recognizing the Heisenberg picture field operators and rearranging one finds that:

$$\int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) = \langle \phi | e^{-it_f \boldsymbol{H}/\hbar}\, \hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_k)\, e^{it_i \boldsymbol{H}/\hbar} | \phi \rangle \quad (t_n > t_k) \tag{47}$$

The combination of equations (46) and (47) yields:

$$\int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) = \begin{cases} \langle \phi | e^{-it_f \boldsymbol{H}/\hbar}\, \hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\, e^{it_i \boldsymbol{H}/\hbar} | \phi \rangle & (t_k > t_n) \\ \langle \phi | e^{-it_f \boldsymbol{H}/\hbar}\, \hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_k)\, e^{it_i \boldsymbol{H}/\hbar} | \phi \rangle & (t_n > t_k) \end{cases}$$

$$\int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) = \langle \phi | e^{-it_f \boldsymbol{H}/\hbar}\, T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)] e^{it_i \boldsymbol{H}/\hbar} | \phi \rangle \tag{48}$$

where $T[\cdots]$ is the *time-ordering* operator defined by:

$$T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)] = \begin{cases} \hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n) & (t_k > t_n) \\ \hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_k) & (t_n > t_k) \end{cases}$$

Equation (48) has a similar time-ordering structure as the Feynman propagator (23) we derived in section 1.3. The right-hand side of (48) is very close to the *two-point correlation function* used in quantum field theory, which for a real scalar field ψ is defined as:

$$\Delta(\boldsymbol{x} - \boldsymbol{y}) = \langle 0 | T[\hat{\psi}(\boldsymbol{x})\hat{\psi}(\boldsymbol{y})] | 0 \rangle \tag{49}$$

where the ket $|0\rangle$ represents the vacuum state[2]. To bridge the gap between (48) and (49) we need to project $|\phi\rangle$ onto this vacuum ground state. We start by expanding the state $e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle$ in the basis of energy eigenstates $|n\rangle$ with associated eigenvalues E_n:

$$e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle = \int \langle n | e^{i\boldsymbol{H}t_i/\hbar} | \phi \rangle |n\rangle \, dn \tag{50}$$

Since $|n\rangle$ is an energy eigenstate it is also an eigenstate of $\boldsymbol{H}$ and therefore:

$$e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle = \int \langle n | e^{iE_n t_i/\hbar} | \phi \rangle |n\rangle \, dn = \int \langle n | \phi \rangle e^{iE_n t_i/\hbar} |n\rangle \, dn$$

We now take the limit $t_i \to -\infty(1 - i\varepsilon)$, with $\varepsilon \to 0^+$. It is essentially equivalent to taking the limit where the initial time goes to $-\infty$, only slightly rotated clockwise toward the positive imaginary complex plane by the term $-\infty(-i\varepsilon)$. In this limit we have:

$$\lim_{t_i \to -\infty(1-i\varepsilon)} e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle = \int \langle n | \phi \rangle e^{iE_n(-\infty(1-i\varepsilon))/\hbar} |n\rangle \, dn$$

$$\lim_{t_i \to -\infty(1-i\varepsilon)} e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle = \int \langle n | \phi \rangle e^{(-\infty)\times\varepsilon E_n/\hbar} \times e^{(-\infty)\times iE_n/\hbar} |n\rangle \, dn$$

In the last equation we sum over all the energy eigenstates from $n = 0$ to $n = \infty$. In the limit $t_i \to -\infty(1 - i\varepsilon)$, the term $e^{(-\infty)\times\varepsilon E_n/\hbar}$ decreases rapidly to zero. Specifically, the higher the energy E_n the faster it exponentially decreases to zero. The vacuum ground state at $n = 0$ must, by design, be the state of lowest energy E_0, i.e. $E_n > E_0$ for $n > 0$. Accordingly, in this sum over all the energy levels, we expect that the $n = 0$ will dominate all the others in this limit. Therefore, in the limit $t_i \to -\infty(1 - i\varepsilon)$, the expansion of $e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle$ in the basis of energy eigenstates is reduced to its projection onto the vacuum ground state $|0\rangle$. Equation (50) thus becomes:

$$\lim_{t_i \to -\infty(1-i\varepsilon)} e^{it_i \boldsymbol{H}/\hbar}|\phi\rangle = \langle 0 | \phi \rangle e^{iE_0 t_i/\hbar} |0\rangle$$

[2]The two-point correlation function $\Delta(\boldsymbol{x} - \boldsymbol{y})$ represents the probability amplitude for a particle, or more precisely a quantum field excitation, to propagate between spacetime points $\boldsymbol{x}$ and $\boldsymbol{y}$.

We can apply the same projection onto the vacuum state logic to $\langle\phi|e^{-it_f\boldsymbol{H}/\hbar}$ and find:

$$\lim_{t_f\to+\infty(1-i\varepsilon)}\langle\phi|e^{-it_f\boldsymbol{H}/\hbar}=\langle\phi|0\rangle e^{-iE_0t_f/\hbar}\langle 0|$$

Substituting these two projections into equation (48) yields:

$$\lim_{\substack{t_i\to-\infty(1-i\varepsilon)\\ t_f\to+\infty(1-i\varepsilon)}}\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)=\langle\phi|0\rangle e^{-iE_0t_f/\hbar}\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)]\langle 0|\phi\rangle e^{iE_0t_i/\hbar}|0\rangle$$

$$\lim_{\substack{t_i\to-\infty(1-i\varepsilon)\\ t_f\to+\infty(1-i\varepsilon)}}\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)=\langle\phi|0\rangle\langle 0|\phi\rangle e^{-iE_0(t_f-t_i)/\hbar}\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)]|0\rangle$$

Had we performed the same projection onto the vacuum state to the simpler path integral

$$\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}$$

we would have found:

$$\lim_{\substack{t_i\to-\infty(1-i\varepsilon)\\ t_f\to+\infty(1-i\varepsilon)}}\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}=\langle\phi|0\rangle\langle 0|\phi\rangle e^{-iE_0(t_f-t_i)/\hbar}$$

Taking the ratio of the previous two results yields:

$$\lim_{\substack{t_i\to-\infty(1-i\varepsilon)\\ t_f\to+\infty(1-i\varepsilon)}}\frac{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)}{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}}=\frac{\langle\phi|0\rangle\langle 0|\phi\rangle e^{-iE_0(t_f-t_i)/\hbar}\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)]|0\rangle}{\langle\phi|0\rangle\langle 0|\phi\rangle e^{-iE_0(t_f-t_i)/\hbar}}$$

$$\lim_{\substack{t_i\to-\infty(1-i\varepsilon)\\ t_f\to+\infty(1-i\varepsilon)}}\frac{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)}{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}}=\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)]|0\rangle$$

And finally, since $\varepsilon\to 0^+$ we can write:

$$\lim_{\substack{t_i\to-\infty\\ t_f\to+\infty}}\frac{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}\,\psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)}{\int\mathcal{D}[\psi(\boldsymbol{x})]e^{iS[\psi(\boldsymbol{x})]/\hbar}}=\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)]|0\rangle \tag{51}$$

The right-hand side matches the definition (49) of the two-point correlation function for a quantum field, which proves that they can be computed from the path integral formulation of quantum field theory. Also note that, in the limits $t_i\to-\infty$ and $t_f\to+\infty$, the action in (51) spans the entire spacetime:

$$S[\psi(x)]=\lim_{\substack{t_i\to-\infty\\ t_f\to+\infty}}\int_{t_i}^{t_f}dt\int\mathcal{L}(\psi,\partial_\mu\psi)\,d^3x=\int\mathcal{L}(\psi,\partial_\mu\psi)\,d^4x$$

The result in (51) can easily be generalized to an arbitrary number of operators to get the n-point correlation function:

$$\lim_{\substack{t_i\to-\infty\\ t_f\to+\infty}}\frac{\int\mathcal{D}[\psi(\boldsymbol{x})]\,e^{iS[\psi(\boldsymbol{x})]/\hbar}\psi(\boldsymbol{x}_1)\times\cdots\times\psi(\boldsymbol{x}_n)}{\int\mathcal{D}[\psi(\boldsymbol{x})]\,e^{iS[\psi(\boldsymbol{x})]/\hbar}}=\langle 0|T[\hat{\psi}(\boldsymbol{x}_1)\cdots\hat{\psi}(\boldsymbol{x}_n)]|0\rangle \tag{52}$$

In the remainder of this section we will introduce the concept of *generating functionals* and their applications to path integrals of quantum fields.

3.3 Functionals and functional derivatives

A function g takes a number x and returns another number $g(x)$. A *functional* F takes a function g and returns a number $F[g(x)]$. The action of Lagrangian mechanics

$$S[x(t)] = \int \mathcal{L}(x, \dot{x})dt$$

is an example of a functional that takes a function, the trajectory $x(t)$, and returns a number, the action. The derivative of a function g is given by:

$$\frac{dg}{dx} = \lim_{\varepsilon \to 0} \frac{g(x+\varepsilon) - g(x)}{\varepsilon}$$

The derivative of a functional F, or *functional derivative*, is defined similarly as:

$$\frac{\delta F[g(x)]}{\delta g(y)} = \lim_{\varepsilon \to 0} \frac{F[g(x) + \varepsilon\delta(x-y)] - F[g(x)]}{\varepsilon}$$

where $\delta(x-y)$ is Dirac's delta function. Applied to the identity functional $F[g(x)] = g(x)$, the functional derivative gives:

$$\frac{\delta g(x)}{\delta g(y)} = \lim_{\varepsilon \to 0} \frac{g(x) + \varepsilon\delta(x-y) - g(x)}{\varepsilon}$$

$$\frac{\delta g(x)}{\delta g(y)} = \delta(x-y) \tag{53}$$

As an example, the functional derivative of

$$F[g(x)] = \int g(x)h(x)dx$$

is given by:

$$\frac{\delta F[g(x)]}{\delta g(y)} = \frac{\delta}{\delta g(y)} \int g(x)h(x)\,dx$$

$$\frac{\delta F[g(x)]}{\delta g(y)} = \int \frac{\delta g(x)}{\delta g(y)} h(x)\,dx$$

$$\frac{\delta F[g(x)]}{\delta g(y)} = \int \delta(x-y)h(x)\,dx$$

$$\frac{\delta F[g(x)]}{\delta g(y)} = h(y)$$

3.4 The generating functional

We define the *generating functional* of a quantum field $\psi(\boldsymbol{x})$ to be the functional:

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]/\hbar} \tag{54}$$

where $J(\boldsymbol{x})$ is a spacetime function called the *source*, and $S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]$ is the action of the field modified by this source term:

$$S[\psi(\boldsymbol{x}), J(\boldsymbol{x})] = \int \Big(\mathcal{L}(\psi, \partial_\mu \psi) + J(\boldsymbol{x})\psi(\boldsymbol{x})\Big) d^4x \tag{55}$$

Let us first show that the functional derivative of $S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]$ with respect to $J(\boldsymbol{x}_k)$, where $\boldsymbol{x}_k = (t^k, x^k, y^k, z^k)$ is a given spacetime coordinate, returns the field value precisely at $\boldsymbol{x}_k$.

$$\frac{\delta S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]}{\delta J(\boldsymbol{x}_k)} = \int \frac{\delta \mathcal{L}(\psi, \partial_\mu \psi)}{\delta J(\boldsymbol{x}_k)}\, d^4x + \int \frac{\delta \left(J(\boldsymbol{x})\psi(\boldsymbol{x})\right)}{\delta J(\boldsymbol{x}_k)}\, d^4x$$

The Lagrangian does not depend on the source term J, hence the first integral is zero. The field $\psi(\boldsymbol{x})$ also does not depend on J, and therefore we can write:

$$\frac{\delta S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]}{\delta J(\boldsymbol{x}_k)} = \int \frac{\delta J(\boldsymbol{x})}{\delta J(\boldsymbol{x}_k)}\, \psi(\boldsymbol{x})\, d^4x$$

$$\frac{\delta S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]}{\delta J(\boldsymbol{x}_k)} = \int \delta(\boldsymbol{x} - \boldsymbol{x}_k)\psi(\boldsymbol{x})\, d^4x$$

$$\frac{\delta S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]}{\delta J(\boldsymbol{x}_k)} = \psi(\boldsymbol{x}_k) \tag{56}$$

which is what we set out to prove. Now we turn to the functional derivative of $Z[J]$ with respect to $J(\boldsymbol{x}_k)$:

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)} = \int \mathcal{D}[\psi(\boldsymbol{x})] \frac{\delta}{\delta J(\boldsymbol{x}_k)} \left(e^{iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]/\hbar}\right)$$

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)} = \frac{i}{\hbar} \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]/\hbar}\, \frac{\delta S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]}{\delta J(\boldsymbol{x}_k)}$$

We already know what the derivative of $S[\psi(\boldsymbol{x}), J(\boldsymbol{x})]$ with respect to $J(\boldsymbol{x}_k)$ is from (56), which yields:

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)} = \frac{i}{\hbar} \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_k) \tag{57}$$

Calculating the second functional derivative of (57) with respect to $J(\boldsymbol{x}_n)$ gives:

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_k)} = \left(\frac{i}{\hbar}\right)^2 \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k)$$

We now evaluate this expression for $J = 0$:

$$\left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_k)}\right|_{J=0} = \left(\frac{i}{\hbar}\right)^2 \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x}), 0]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) \tag{58}$$

$$(-i\hbar)^2 \left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_k)}\right|_{J=0} = \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}\, \psi(\boldsymbol{x}_n)\psi(\boldsymbol{x}_k) \tag{59}$$

where in the last line we have used equation (55) to identify $S[\psi(\boldsymbol{x}), 0]$ as the classical field action $S[\psi(\boldsymbol{x})]$. Finally, substituting this result into equation (51) yields:

$$\frac{(-i\hbar)^2}{\int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}} \left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_k)}\right|_{J=0} = \langle 0 | T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)] | 0 \rangle$$

$$\frac{(-i\hbar)^2}{Z[0]} \left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_k)}\right|_{J=0} = \langle 0 | T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)] | 0 \rangle \tag{60}$$

where we have defined $Z[0] = \int \mathcal{D}[\psi(\boldsymbol{x})] e^{iS[\psi(\boldsymbol{x})]/\hbar}$ to be the normalization constant. Equation (60) shows that the two-point correlation function of a quantum field can be calculated by

taking successive functional derivatives of the field's generating functional. Again, this result can easily be generalized to a n-point correlation function:

$$\frac{(-i\hbar)^n}{Z[0]} \frac{\delta^n Z[J]}{\delta J(\boldsymbol{x}_1)\cdots\delta J(\boldsymbol{x}_n)}\bigg|_{J=0} = \langle 0|T[\hat{\psi}(\boldsymbol{x}_1)\cdots\hat{\psi}(\boldsymbol{x}_n)]|0\rangle \tag{61}$$

This remarkable result means that everything we need to know about the dynamics of a quantum field is enclosed in its generating functional. The ad-hoc source term $J(\boldsymbol{x})$ was introduced merely as a temporary tool to *shake out* field values at designated spacetime points from the exponential $e^{iS[\psi(\boldsymbol{x}),J(\boldsymbol{x})]/\hbar}$ (cf. (56)). Once we got these, this tool was discarded when we evaluated the result for $J = 0$ (cf. (58)). Calculating n-point correlation functions in the canonical field quantization formalism can be a tedious task because one has to keep track of all the commutation and anti-commutation relations between various field operators evaluated at different spacetime points. The power of the generating functional approach is that it manipulates field values, which always commute [9].

3.5 Wick's theorem

Another remarkable aspect of the path integral quantization of fields is that we can find Wick's theorem by taking successive functional derivatives of the generating functional. We will consider the 4-point correlation function $\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_p)\hat{\psi}(\boldsymbol{x}_q)]|0\rangle$ to see how this unfolds. Equation (61) indicates that we have to compute the fourth functional derivative:

$$\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_p)\hat{\psi}(\boldsymbol{x}_q)]|0\rangle = \frac{(-i)^4}{Z[0]} \frac{\delta^4 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_p)\delta J(\boldsymbol{x}_q)}\bigg|_{J=0} \tag{62}$$

Let us tackle each functional derivative in turn (the calculations are tedious yet not difficult):

$$\frac{\delta^3}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_p)}\left[\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_q)}\right] = \frac{\delta^3}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)\delta J(\boldsymbol{x}_p)}\left[(-i)Z[J]\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right]$$

$$
\begin{aligned}
=&\frac{\delta^2}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\left[(-i)\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_p)}\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x+(-i)Z[J]G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\right]\\
=&\frac{\delta^2}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\Big[(-i)^2Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\\
&\times\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)+(-i)Z[J]G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big]\\
=&\frac{\delta}{\delta J(\boldsymbol{x}_k)}\Big[(-i)^2\frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\\
&\times\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\Big)+(-i)\frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(Z[J]G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big)\Big]\\
=&\frac{\delta}{\delta J(\boldsymbol{x}_k)}\Big[(-i)^2\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)}\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\\
&\times\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+(-i)^2Z[J]G_F(\boldsymbol{x}_p-\boldsymbol{x}_n)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+(-i)^2Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+(-i)\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)}G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big]\\
=&\frac{\delta}{\delta J(\boldsymbol{x}_k)}\Big[(-i)^3Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)\\
&\times\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+(-i)^2Z[J]G_F(\boldsymbol{x}_p-\boldsymbol{x}_n)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+(-i)^2Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+(-i)^2Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&= (-i)^3\Big[\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)\\
&\times\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+ Z[J]G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+ Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_p-\boldsymbol{x}_k)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+ Z[J]\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_k)\Big]\\
&+ (-i)^2\Big[\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}G_F(\boldsymbol{x}_p-\boldsymbol{x}_n)\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_q-\boldsymbol{x})\,d^4x\right)\\
&+ Z[J]G_F(\boldsymbol{x}_p-\boldsymbol{x}_n)G_F(\boldsymbol{x}_q-\boldsymbol{x}_k)\\
&+ \frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_p-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+ Z[J]G_F(\boldsymbol{x}_p-\boldsymbol{x}_k)G_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+ \frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\left(\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right)G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\\
&+ Z[J]G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&= (-i)^2\Big(Z[0]G_F(\boldsymbol{x}_p-\boldsymbol{x}_n)G_F(\boldsymbol{x}_q-\boldsymbol{x}_k)+Z[0]G_F(\boldsymbol{x}_p-\boldsymbol{x}_k)G_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+ Z[0]G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)G_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big)
\end{aligned}
$$

$$
\begin{aligned}
&= Z[0]\Big(iG_F(\boldsymbol{x}_p-\boldsymbol{x}_n)iG_F(\boldsymbol{x}_q-\boldsymbol{x}_k)+iG_F(\boldsymbol{x}_p-\boldsymbol{x}_k)iG_F(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+ iG_F(\boldsymbol{x}_n-\boldsymbol{x}_k)iG_F(\boldsymbol{x}_q-\boldsymbol{x}_p)\Big)
\end{aligned}
$$

We finally substitute this result into (62) to get:

$$
\begin{aligned}
\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_p)\hat{\psi}(\boldsymbol{x}_q)]|0\rangle &= \Delta(\boldsymbol{x}_p-\boldsymbol{x}_n)\Delta(\boldsymbol{x}_q-\boldsymbol{x}_k)+\Delta(\boldsymbol{x}_p-\boldsymbol{x}_k)\Delta(\boldsymbol{x}_q-\boldsymbol{x}_n)\\
&+ \Delta(\boldsymbol{x}_n-\boldsymbol{x}_k)\Delta(\boldsymbol{x}_q-\boldsymbol{x}_p)
\end{aligned}
$$

This matches Wick's theorem which states that the 4-point correlation function

$$\langle 0|T[\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_p)\hat{\psi}(\boldsymbol{x}_q)]|0\rangle$$

can be written as the sum of all possible contractions of $\hat{\psi}(\boldsymbol{x}_k)\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_p)\hat{\psi}(\boldsymbol{x}_q)$.

4 Path integral quantization of the Klein-Gordon scalar field

4.1 Deriving the generating functional

The relativistic equation of motion for a free scalar field ψ of mass m is given by the Klein-Gordon equation [10, 11]:

$$\left(\partial^2 + m^2\right)\psi(\boldsymbol{x}) = 0 \Leftrightarrow \left(\partial_\mu\partial^\mu + m^2\right)\psi(\boldsymbol{x}) = 0 \tag{63}$$

The Klein-Gordon Lagrangian is:

$$\mathcal{L}(\psi, \partial_\mu\psi) = \frac{1}{2}\left(\partial_\mu\psi(\boldsymbol{x})\right)^2 - \frac{1}{2}m^2\psi(\boldsymbol{x})^2 \tag{64}$$

This is because applying the Euler-Lagrange equation

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\right) - \frac{\partial\mathcal{L}}{\partial\psi} = 0$$

to (64) gives the Klein-Gordon equation (63):

$$\partial_\mu\left(\frac{\partial}{\partial(\partial_\mu\psi)}\left(\frac{1}{2}\left(\partial_\mu\psi\right)^2 - \frac{1}{2}m^2\psi(\boldsymbol{x})^2\right)\right) - \frac{\partial}{\partial\psi}\left(\frac{1}{2}\left(\partial_\mu\psi\right)^2 - \frac{1}{2}m^2\psi(\boldsymbol{x})^2\right) = 0$$

$$\partial_\mu\left(\frac{\partial}{\partial(\partial_\mu\psi)}\left(\frac{1}{2}(\partial^\mu\psi)(\partial_\mu\psi)\right)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$\partial_\mu\left(\frac{\partial}{\partial(\partial_\mu\psi)}\left(\frac{1}{2}(\eta^{\mu\nu}\partial_\nu\psi)(\partial_\mu\psi)\right)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$\partial_\mu\left(\frac{\eta^{\mu\nu}}{2}\left(\frac{\partial(\partial_\nu\psi)}{\partial(\partial_\mu\psi)}\partial_\mu\psi + \partial_\nu\psi\frac{\partial(\partial_\mu\psi)}{\partial(\partial_\mu\psi)}\right)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$\partial_\mu\left(\frac{\eta^{\mu\nu}}{2}(\delta^\mu_\nu\partial_\mu\psi + \partial_\nu\psi)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$\partial_\mu\left(\frac{1}{2}(\eta^{\mu\nu}\partial_\nu\psi + \eta^{\mu\nu}\partial_\nu\psi)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$\partial_\mu\left(\frac{1}{2}(\partial^\mu\psi + \partial^\mu\psi)\right) + m^2\psi(\boldsymbol{x}) = 0$$

$$(\partial_\mu\partial^\mu + m^2)\psi(\boldsymbol{x}) = 0$$

■

The generating functional for the Klein-Gordon field is given by (we will work with natural units where $c = \hbar = 1$):

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})]\exp\left(iS[\psi(\boldsymbol{x}), J(\boldsymbol{x})]\right)$$

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})]\exp\left(i\int\left(\mathcal{L}(\psi, \partial_\mu\psi) + J(\boldsymbol{x})\psi(\boldsymbol{x})\right)d^4x\right)$$

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})]\exp\left(i\int\left(\frac{1}{2}\left(\partial_\mu\psi\right)^2 - \frac{1}{2}m^2\psi(\boldsymbol{x})^2 + J(\boldsymbol{x})\psi(\boldsymbol{x})\right)d^4x\right)$$

First, we tackle the term $\left(\partial_\mu\psi\right)^2$ with an integration by parts:

$$\int\left(\partial_\mu\psi(\boldsymbol{x})\right)^2 d^4x = \int\left(\partial_\mu\psi(\boldsymbol{x})\right)\left(\partial^\mu\psi(\boldsymbol{x})\right)d^4x$$

$$\int\left(\partial_\mu\psi(\boldsymbol{x})\right)^2 d^4x = \int\partial_\mu\big(\psi(\boldsymbol{x})\partial^\mu\psi(\boldsymbol{x})\big)d^4x - \int\psi(\boldsymbol{x})\partial^2\psi(\boldsymbol{x})\,d^4x$$

The field is assumed to vanish at the boundary of spacetime, i.e. $\psi(\boldsymbol{x}) = 0$ for $|\boldsymbol{x}| \to \infty$, and we can therefore write:

$$\int \partial_\mu\big(\psi(\boldsymbol{x})\partial^\mu\psi(\boldsymbol{x})\big)d^4x = 0 \Rightarrow \int (\partial_\mu\psi(\boldsymbol{x}))^2 d^4x = -\int \psi(\boldsymbol{x})\partial^2\psi(\boldsymbol{x})\, d^4x$$

$$\Rightarrow Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})] \exp\left(i\int\left(-\frac{1}{2}\psi(\boldsymbol{x})\partial^2\psi(\boldsymbol{x}) - \frac{1}{2}m^2\psi(\boldsymbol{x})^2 + J(\boldsymbol{x})\psi(\boldsymbol{x})\right)d^4x\right)$$

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})] \exp\left(\int\left(\frac{i}{2}\psi(\boldsymbol{x})\left[-(\partial^2 + m^2)\right]\psi(\boldsymbol{x}) + iJ(\boldsymbol{x})\psi(\boldsymbol{x})\right)d^4x\right) \tag{65}$$

Our next challenge is to evaluate this infinite-dimensional functional Gaussian integral.

4.2 Functional Gaussian integrals

We would like to generalize the Gaussian integral

$$\int e^{-\frac{1}{2}ax^2}dx$$

in which x is a scalar, to a functional Gaussian integral

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T -\frac{1}{2}x(t)\hat{A}x(t)dt\right) \tag{66}$$

where $x(t)$ is a trajectory whose boundary values at $x(0)$ and $x(T)$ are fixed, and $\hat{A}$ is a linear differential operator. For the Klein-Gordon field, we will set $\hat{A} = -(\partial^2 + m^2)$ (cf. (65)). The idea is to split the interval $[0, T]$ into N time-slices of infinitesimal length δt, and to represent the trajectory $x(t)$ as an $(N+1)$-dimensional vector

$$\mathbf{x} = \begin{pmatrix} x_0 = x(0) \\ \vdots \\ x_n = x(n\delta t) \\ \vdots \\ x_N = x(T) \end{pmatrix}$$

where the boundary components x_0 and x_N are fixed. With this representation, the exponent in (66) becomes:

$$\int_0^T -\frac{1}{2}x(t)\hat{A}x(t)dt \;\longrightarrow\; -\frac{1}{2}\,\mathbf{x}^T\boldsymbol{A}\,\mathbf{x} = -\frac{1}{2}\sum_{k,i} x_i A_{ik} x_k$$

where $\mathbf{x}^T$ is the transposed vector $\mathbf{x}$, and $\boldsymbol{A}$ is an $(N+1)\times(N+1)$ matrix representation of the linear operator $\hat{A}$. Note that we have absorbed the δt factor inside the definition of $\boldsymbol{A}$ when we transitioned from a continuous integral $\int dt$ to the discrete sum $\sum \delta t$. With these definitions, the discretized version of the path integral in (66) becomes:

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\,\mathbf{x}^T\boldsymbol{A}\,\mathbf{x}\right) \tag{67}$$

We will consider a trajectory such that $x(0) = x(T) = 0$, which is consistent with the fact that we are interested in vacuum-to-vacuum transition amplitudes where the field is zero everywhere

at the boundaries (cf. (51)). Since x_0 and x_N are zero the trajectory vector is essentially reduced to the $(N-1)$-dimensional vector:

$$\mathbf{x} = \begin{pmatrix} x_1 = x(\delta t) \\ \vdots \\ x_n = x(n\delta t) \\ \vdots \\ x_{N-1} = x(T - \delta t) \end{pmatrix}$$

and $\boldsymbol{A}$ is an $(N-1) \times (N-1)$ matrix. For non-pathological linear differential operators, the matrix representation will be symmetric and therefore diagonalizable. Consider the operator $(\frac{d^2}{dt^2} + m^2)$ as an example. Its discrete approximation evaluated at $t = t_n$ is given by:

$$\left(\frac{d^2}{dt^2} + m^2\right) x(t)\Big|_{t=t_n} \rightarrow \frac{1}{\delta t}\left(\frac{x_{n+1} - x_n}{\delta t} - \frac{x_n - x_{n-1}}{\delta t}\right) + m^2 x_n = \frac{x_{n+1} - 2x_n + x_{n-1}}{(\delta t)^2} + m^2 x_n$$

Because we have decided to absorb the integration measure δt into the definition of its matrix representation, i.e. $\int \hat{A} dt \rightarrow \sum \boldsymbol{A}$ instead of $\int \hat{A} dt \rightarrow \sum \boldsymbol{A}\delta t$, we have to multiply the previous expression by δt. The symmetric matrix associated with the operator $(\frac{d^2}{dt^2} + m^2)$ is therefore:

$$\frac{1}{\delta t}\begin{pmatrix} m^2(\delta t)^2 - 2 & 1 & 0 & \cdots \\ 1 & m^2(\delta t)^2 - 2 & 1 & \cdots \\ 0 & 1 & m^2(\delta t)^2 - 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Let $\boldsymbol{D}$ be the diagonal matrix representation of the Klein-Gordon operator $\boldsymbol{A}$ such that

$$\boldsymbol{A} = \boldsymbol{O}\boldsymbol{D}\boldsymbol{O}^T$$

where $\boldsymbol{O}$ is an orthogonal matrix. The diagonal elements of $\boldsymbol{D}$, which are its only non-zero components, are its eigenvalues denoted D_{ii}. Inserting this diagonal form of $\boldsymbol{A}$ in (67) yields:

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}\right) = \int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{O}\boldsymbol{D}\boldsymbol{O}^T \mathbf{x}\right)$$

After proceeding with the change of variables $\mathbf{z} = \boldsymbol{O}^T\mathbf{x} \Leftrightarrow \mathbf{z}^T = \mathbf{x}^T\boldsymbol{O}$ we obtain:

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}\right) = \int dz_1 \cdots dz_{N-1} \exp\left(-\frac{1}{2}\mathbf{z}^T \boldsymbol{D}\, \mathbf{z}\right) |\det(\boldsymbol{O})|$$

where $|\det(\boldsymbol{O})|$ is the Jacobian associated with this change of variables. But since $\boldsymbol{O}$ is an orthogonal matrix we have $|\det(\boldsymbol{O})| = 1$, and therefore:

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}\right) = \int dz_1 \cdots dz_{N-1} \exp\left(-\frac{1}{2}\mathbf{z}^T \boldsymbol{D}\, \mathbf{z}\right)$$

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}\right) = \int dz_1 \cdots dz_{N-1} \exp\left(-\frac{1}{2}\sum_{i=1}^{N-1}(z_i)^2 D_{ii}\right)$$

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}\right) = \prod_{i=1}^{N-1} \int dz_i \exp\left(-\frac{1}{2}(z_i)^2 D_{ii}\right)$$

Each integral in the product is a Gaussian integral of the form

$$\int dz_i \exp\left(-\frac{1}{2}(z_i)^2 D_{ii}\right) = \sqrt{\frac{2\pi}{D_{ii}}}$$

which yields:

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\,\mathbf{x}^T \boldsymbol{A}\,\mathbf{x}\right) = \prod_{i=1}^{N-1} \sqrt{\frac{2\pi}{D_{ii}}}$$

$$\int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\,\mathbf{x}^T \boldsymbol{A}\,\mathbf{x}\right) = \sqrt{\frac{(2\pi)^{N-1}}{\det(\boldsymbol{D})}}$$

where $\det(\boldsymbol{D}) = D_{11} \times D_{22} \times \cdots$ is the determinant of the diagonal matrix $\boldsymbol{D}$. On the other hand we have:

$$\boldsymbol{A} = \boldsymbol{O}\boldsymbol{D}\boldsymbol{O}^T$$

$$\det(\boldsymbol{A}) = \det(\boldsymbol{O})\det(\boldsymbol{D})\det(\boldsymbol{O}^T) = \det(\boldsymbol{D})\det(\boldsymbol{O})\det(\boldsymbol{O}^T)$$

$$\det(\boldsymbol{A}) = \det(\boldsymbol{D})$$

$$\Rightarrow \int dx_1 \cdots dx_{N-1} \exp\left(-\frac{1}{2}\,\mathbf{x}^T \boldsymbol{A}\,\mathbf{x}\right) = \sqrt{\frac{(2\pi)^{N-1}}{\det(\boldsymbol{A})}}$$

where we have used the property of orthogonal matrices that $\det(\boldsymbol{O}) = \pm 1$. Thanks to this result, we can write the functional Gaussian integral in (66) as:

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T -\frac{1}{2} x(t)\hat{A}x(t)dt\right) = \lim_{N\to\infty} \sqrt{\frac{(2\pi)^{N-1}}{\det(\boldsymbol{A})}} \tag{68}$$

We can generalize this result to the Gaussian integral

$$\int \exp\left(-\frac{1}{2}ax^2 + bx\right) dx = \exp\left(\frac{b^2}{2a}\right)\sqrt{\frac{2\pi}{a}}$$

whose functional integral is given by:

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T \left(-\frac{1}{2} x(t)\hat{A}x(t) + b(t)x(t)\right) dt\right) = \lim_{N\to\infty} \sqrt{\frac{(2\pi)^{N-1}}{\det(\boldsymbol{A})}} \exp\left(\frac{1}{2}\,\mathbf{b}^T \boldsymbol{A}^{-1}\mathbf{b}\right)$$

$\boldsymbol{A}^{-1}$ is the inverse matrix associated with the differential operator $\hat{A}$, such that $\boldsymbol{A}\boldsymbol{A}^{-1} = \mathbb{I}$. After redefining $\hat{A} \to -i\hat{A}$ and $b(t) \to ib(t)$, this becomes:

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T \left(\frac{i}{2} x(t)\hat{A}x(t) + ib(t)x(t)\right) dt\right) = \lim_{N\to\infty} \sqrt{\frac{(2\pi)^{N-1}}{\det(-i\boldsymbol{A})}} \exp\left(\frac{1}{2}\, i\mathbf{b}^T(-i\boldsymbol{A})^{-1} i\mathbf{b}\right)$$

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T \left(\frac{i}{2} x(t)\hat{A}x(t) + ib(t)x(t)\right) dt\right) = \lim_{N\to\infty} \sqrt{\frac{(2\pi)^{N-1}}{(-i)^{N-1}\det(\boldsymbol{A})}} \exp\left(\frac{-i}{2}\,\mathbf{b}^T \boldsymbol{A}^{-1}\mathbf{b}\right)$$

$$\int \mathcal{D}[x(t)] \exp\left(\int_0^T \left(\frac{i}{2} x(t)\hat{A}x(t) + ib(t)x(t)\right) dt\right) = \lim_{N\to\infty} \sqrt{\frac{(2\pi i)^{N-1}}{\det(\boldsymbol{A})}} \exp\left(\frac{-i}{2}\,\mathbf{b}^T \boldsymbol{A}^{-1}\mathbf{b}\right)$$

At last, we can use this result to evaluate the generating functional of the Klein-Gordon field that we found in (65):

$$Z[J] = \int \mathcal{D}[\psi(\boldsymbol{x})] \exp\left(\int \left(\frac{i}{2}\psi(\boldsymbol{x})\left[-(\partial^2 + m^2)\right]\psi(\boldsymbol{x}) + iJ(\boldsymbol{x})\psi(\boldsymbol{x})\right) d^4x\right)$$

Substituting $\hat{A}$ with $-(\partial^2 + m^2)$ and the function b with the source term J, we find:

$$Z[J] = \lim_{N\to\infty} \sqrt{\frac{(2\pi i)^{N-1}}{\det(\boldsymbol{A})}} \exp\left(-\frac{i}{2}\,\mathbf{J}^T \boldsymbol{A}^{-1}\mathbf{J}\right) \tag{69}$$

where $\boldsymbol{A}$ is the matrix representation of the operator $-(\partial^2 + m^2)$. In the limit $N \to \infty$, the matrix product in the exponent becomes a continuous integral:

$$\mathbf{J}^T \boldsymbol{A}^{-1}\mathbf{J} = \sum_{i,j} J_i (\boldsymbol{A}^{-1})_{ij} J_j \;\longrightarrow\; \int J(\boldsymbol{x}) G(\boldsymbol{x},\boldsymbol{y}) J(\boldsymbol{y})\, d^4x\, d^4y$$

where we have defined $G(\boldsymbol{x},\boldsymbol{y})$ to be the continuous limit of the $\boldsymbol{A}^{-1}$ matrix when $N \to \infty$. Therefore, in the limit $N \to \infty$, the relation $\boldsymbol{A}\boldsymbol{A}^{-1} = \mathbb{I}$ becomes:

$$\boldsymbol{A}\boldsymbol{A}^{-1} = \mathbb{I} \;\longrightarrow\; -(\partial^2 + m^2) G(\boldsymbol{x},\boldsymbol{y}) = \delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$$

where $\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$ is the 4-dimensional delta function. Looking back at equation (2), this expression matches the definition of the Green function of the operator $(\partial^2 + m^2)$. We choose the Feynman version of the Green function, which we know is symmetric and depends only on the spacetime separation $(\boldsymbol{x}-\boldsymbol{y})$ (cf. 1.3). Accordingly, we can rewrite equation (69) as:

$$Z[J] = \mathcal{N} \exp\left(-\frac{i}{2}\int J(\boldsymbol{x}) G_F(\boldsymbol{x}-\boldsymbol{y}) J(\boldsymbol{y})\, d^4x\, d^4y\right) \tag{70}$$

$$\text{with } \mathcal{N} = \lim_{N\to\infty} \sqrt{\frac{(2\pi i)^{N-1}}{\det(-(\partial^2 + m^2))}}$$

where $G_F(\boldsymbol{x}-\boldsymbol{y})$ is the Feynman propagator of the Klein-Gordon field. As usual, we don't have to worry about the diverging constant $\mathcal{N}$ because it will be factored out during the normalization process.

In this section we have shown that the generating functional of a quantum field contains its propagator. As we will see in the next section, this explains the connection between two-point correlation functions in quantum field theory and propagators.

4.3 The two-point correlation function of the Klein-Gordon field

In (60) we found that the two-point correlation function of a quantum field ψ can be calculated as the second functional derivative:

$$\Delta(\boldsymbol{x}_n - \boldsymbol{x}_k) = \langle 0|T[\hat{\psi}(\boldsymbol{x}_n)\hat{\psi}(\boldsymbol{x}_k)]|0\rangle = \frac{(-i)^2}{Z[0]}\,\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\bigg|_{J=0} \tag{71}$$

We start with the functional derivative with respect to $J(\boldsymbol{x}_n)$:

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)} = \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big[\mathcal{N}\exp\left(-\frac{i}{2}\int J(\boldsymbol{x}) G_F(\boldsymbol{x}-\boldsymbol{y}) J(\boldsymbol{y})\, d^4x\, d^4y\right)\Big]$$

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)} = \mathcal{N}\exp\left(-\frac{i}{2}\int J(\boldsymbol{x}) G_F(\boldsymbol{x}-\boldsymbol{y}) J(\boldsymbol{y})\, d^4x\, d^4y\right) \times \left(-\frac{i}{2}\right)\frac{\delta}{\delta J(\boldsymbol{x}_n)}\int J(\boldsymbol{x}) G_F(\boldsymbol{x}-\boldsymbol{y}) J(\boldsymbol{y})\, d^4x\, d^4y$$

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)} = Z[J]\left(-\frac{i}{2}\right)\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x}) G_F(\boldsymbol{x}-\boldsymbol{y}) J(\boldsymbol{y})\Big) d^4x\, d^4y \tag{72}$$

We now apply the product rule to the functional derivative inside the integrand:

$$\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\Big)d^4x\,d^4y = \int \frac{\delta J(\boldsymbol{x})}{\delta J(\boldsymbol{x}_n)}G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\,d^4x\,d^4y + \int J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})\frac{\delta J(\boldsymbol{y})}{\delta J(\boldsymbol{x}_n)}\,d^4x\,d^4y$$

$$\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\Big)d^4x\,d^4y = \int \delta(\boldsymbol{x}-\boldsymbol{x}_n)G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\,d^4x\,d^4y + \int J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})\delta(\boldsymbol{y}-\boldsymbol{x}_n)\,d^4x\,d^4y$$

$$\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\Big)d^4x\,d^4y = \int J(\boldsymbol{y})G_F(\boldsymbol{x}_n-\boldsymbol{y})\,d^4y + \int J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{x}_n)\,d^4x$$

$$\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\Big)d^4x\,d^4y = \int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x + \int J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{x}_n)\,d^4x$$

$$\int \frac{\delta}{\delta J(\boldsymbol{x}_n)}\Big(J(\boldsymbol{x})G_F(\boldsymbol{x}-\boldsymbol{y})J(\boldsymbol{y})\Big)d^4x\,d^4y = 2\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x$$

Replacing in (72) yields:

$$\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)} = (-i)Z[J]\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x \tag{73}$$

Moving on to the second derivative with respect to $J(\boldsymbol{x}_k)$:

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)} = \frac{\delta}{\delta J(\boldsymbol{x}_k)}\left[\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_n)}\right]$$

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)} = \frac{\delta}{\delta J(\boldsymbol{x}_k)}\left[(-i)Z[J]\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x\right]$$

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)} = (-i)\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x + (-i)Z[J]\int \frac{\delta J(\boldsymbol{x})}{\delta J(\boldsymbol{x}_k)}G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x$$

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)} = (-i)\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x + (-i)Z[J]\int \delta(\boldsymbol{x}-\boldsymbol{x}_k)G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x$$

$$\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)} = (-i)\frac{\delta Z[J]}{\delta J(\boldsymbol{x}_k)}\int J(\boldsymbol{x})G_F(\boldsymbol{x}_n-\boldsymbol{x})\,d^4x + (-i)Z[J]G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)$$

Now we evaluate this expression for $J=0$:

$$\left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\right|_{J=0} = (-i)Z[0]G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)$$

$$\frac{(-i)^2}{Z[0]}\left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\right|_{J=0} = (-i)^2(-i)G_F(\boldsymbol{x}_n-\boldsymbol{x}_k)$$

$$\frac{(-i)^2}{Z[0]}\left.\frac{\delta^2 Z[J]}{\delta J(\boldsymbol{x}_k)\delta J(\boldsymbol{x}_n)}\right|_{J=0} = iG_F(\boldsymbol{x}_n-\boldsymbol{x}_k)$$

Substituting this result into equation (71) gives us the expression for the two-point correlation function:

$$\Delta(\boldsymbol{x}_n-\boldsymbol{x}_k) = iG_F(\boldsymbol{x}_n-\boldsymbol{x}_k) \tag{74}$$

As advertised, the two-point correlation function is simply i times the Feynman propagator. We can show that the Green function for the Klein-Gordon field is

$$G(\boldsymbol{x}-\boldsymbol{y}) = \frac{1}{(2\pi)^4}\int \frac{e^{-i\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})}}{\boldsymbol{p}^2-m^2}\,d^4p \tag{75}$$

by checking that it satisfies the definition of the Green function:

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\left(\partial^2\frac{1}{(2\pi)^4}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p + m^2\frac{1}{(2\pi)^4}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\frac{1}{(2\pi)^4}\left(\partial_\mu\partial^\mu\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p + m^2\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\frac{1}{(2\pi)^4}\left(\left(\frac{\partial^2}{\partial t^2}-\nabla^2\right)\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p + m^2\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\frac{1}{(2\pi)^4}\left(\int \frac{(ip^t)^2-(i|\vec{p}|)^2}{\boldsymbol{p}^2-m^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p + m^2\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\frac{1}{(2\pi)^4}\left(\int \frac{-(p^t)^2+\vec{p}^2}{\boldsymbol{p}^2-m^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p + m^2\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = -\frac{1}{(2\pi)^4}\left(\int \frac{-\boldsymbol{p}^2}{\boldsymbol{p}^2-m^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p + m^2\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2-m^2}\,d^4p\right)$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = \frac{1}{(2\pi)^4}\int \frac{\boldsymbol{p}^2-m^2}{\boldsymbol{p}^2-m^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = \frac{1}{(2\pi)^4}\int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p$$

$$-(\partial^2+m^2)G(\boldsymbol{x}-\boldsymbol{y}) = \delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$$

■

To find the expression for the Feynman propagator we simply have to follow the Feynman contour prescription described in section 1.3:

$$G_F(\boldsymbol{x}-\boldsymbol{y}) = \lim_{\varepsilon\to 0^+}\frac{1}{(2\pi)^4}\int \frac{e^{-i\boldsymbol{p}\cdot(\boldsymbol{x}-\boldsymbol{y})}}{\boldsymbol{p}^2-m^2+i\varepsilon}\,d^4p \tag{76}$$

Finally, substituting this result in (74) gives us the two-point correlation function of the Klein-Gordon field:

$$\Delta(\boldsymbol{x}_n-\boldsymbol{x}_k) = \lim_{\varepsilon\to 0^+}\frac{1}{(2\pi)^4}\int \frac{ie^{-i\boldsymbol{p}\cdot(\boldsymbol{x}_n-\boldsymbol{x}_k)}}{\boldsymbol{p}^2-m^2+i\varepsilon}\,d^4p \tag{77}$$

5 Path integral quantization of the electromagnetic gauge field

5.1 A naive generating functional

The Lagrangian of the free electromagnetic field is given by:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}(\boldsymbol{x})F^{\mu\nu}(\boldsymbol{x}) \quad \text{with} \quad F_{\mu\nu} = \partial_\mu A_\nu(\boldsymbol{x}) - \partial_\nu A_\mu(\boldsymbol{x}) \tag{78}$$

$\boldsymbol{A}$ is the electromagnetic 4-potential whose components are $A^\mu = (\mathcal{V}, \vec{\mathcal{A}})$, where $\mathcal{V}$ is the scalar electric potential and $\vec{\mathcal{A}}$ is the vector magnetic potential. Applying the Euler-Lagrange equation to (78) yields the free electromagnetic field equations:

$$\partial_\mu F^{\mu\nu}(\boldsymbol{x}) = 0 \tag{79}$$

After expanding the terms in the Lagrangian one finds:

$$\mathcal{L} = -\frac{1}{4}(\partial_\mu A_\nu(\boldsymbol{x}) - \partial_\nu A_\mu(\boldsymbol{x}))(\partial^\mu A^\nu(\boldsymbol{x}) - \partial^\nu A^\mu(\boldsymbol{x}))$$

$$\mathcal{L} = -\frac{1}{4}\Big(\partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x}) - \partial_\nu A_\mu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) + \partial_\nu A_\mu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\Big)$$

Consistent with Einstein's summation convention, μ and ν are dummy indices and can therefore be freely swapped:

$$\partial_\nu A_\mu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) \to \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x}) \quad \text{and} \quad \partial_\nu A_\mu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x}) \to \partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})$$

$$\Rightarrow \mathcal{L} = -\frac{1}{4}\Big(\partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x}) + \partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})\Big)$$

$$\mathcal{L} = -\frac{1}{2}\Big(\partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\Big)$$

This is the Lagrangian expression we will use to write the generating functional of the electromagnetic field:

$$Z[J] = \int \mathcal{D}[A^\mu(\boldsymbol{x})] \exp\left(i \int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\, d^4x\right)$$

$$Z[J] = \int \mathcal{D}[A^\mu(\boldsymbol{x})] \exp\left(i \int -\frac{1}{2}\left(\partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\right) d^4x + i \int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}) d^4x\right)$$

Notice that the source term J_μ is now a 4-vector representing the electric current density. Expanding and integrating by parts the first integral in the exponent yields:

$$-\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\Big[A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})\Big]_{-\infty}^{+\infty} + \frac{1}{2}\int A_\nu(\boldsymbol{x})\partial_\mu\partial^\mu A^\nu(\boldsymbol{x})\, d^4x$$

$$-\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})\, d^4x = \frac{1}{2}\int A_\nu(\boldsymbol{x})\partial^2 A^\nu(\boldsymbol{x})\, d^4x$$

$$-\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x})\, d^4x = \frac{1}{2}\int \eta_{\mu\nu}A^\mu(\boldsymbol{x})\partial^2 A^\nu(\boldsymbol{x})\, d^4x \tag{80}$$

where $\eta_{\mu\nu}$ is the Minkowski metric, and in the second line we have used the assumption that the field vanishes at the boundaries of the spacetime integration volume, i.e. that $A_\nu(\boldsymbol{x}) = 0$

for $|\boldsymbol{x}| \to \infty$. Integrating the second integral in $Z[J]$ by parts one finds, after reshuffling some indices:

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = \frac{1}{2}\Big[A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\Big]_{-\infty}^{+\infty} - \frac{1}{2}\int A_\nu(\boldsymbol{x})\partial_\mu\partial^\nu A^\mu(\boldsymbol{x})\, d^4x$$

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\int A_\nu(\boldsymbol{x})\partial^\nu\partial_\mu A^\mu(\boldsymbol{x})\, d^4x$$

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\int \eta_{\sigma\nu}A^\sigma(\boldsymbol{x})\partial^\nu\partial_\mu A^\mu(\boldsymbol{x})\, d^4x$$

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\int A^\sigma(\boldsymbol{x})\eta_{\sigma\nu}\partial^\nu\partial_\mu A^\mu(\boldsymbol{x})\, d^4x$$

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\int A^\sigma(\boldsymbol{x})\partial_\sigma\partial_\mu A^\mu(\boldsymbol{x})\, d^4x$$

$$\frac{1}{2}\int \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\, d^4x = -\frac{1}{2}\int A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})\, d^4x \tag{81}$$

Summing (80) and (81) one finds that the integral can be reduced to:

$$\begin{aligned}\int -\frac{1}{2}\Big(\partial_\mu A_\nu(\boldsymbol{x})\partial^\mu A^\nu(\boldsymbol{x}) - \partial_\mu A_\nu(\boldsymbol{x})\partial^\nu A^\mu(\boldsymbol{x})\Big)d^4x& \\ = \frac{1}{2}\int \eta_{\mu\nu}A^\mu(\boldsymbol{x})\partial^2 A^\nu(\boldsymbol{x})\, d^4x - \frac{1}{2}\int A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})\, d^4x& \\ = \frac{1}{2}\int \Big(\eta_{\mu\nu}A^\mu(\boldsymbol{x})\partial^2 A^\nu(\boldsymbol{x}) - A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})\Big)d^4x& \\ = \frac{1}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu} - \partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\, d^4x&\end{aligned}$$

Substituting this result into our expression for $Z[J]$ yields:

$$Z[J] = \int \mathcal{D}[A^\mu(\boldsymbol{x})]\exp\left(\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu} - \partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\, d^4x + i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})d^4x\right) \tag{82}$$

This is the same type of functional Gaussian integral we solved for the Klein-Gordon field in (65). The only difference is that, instead of scalar fields, we are dealing with 4-dimensional vector fields A^μ and J_μ, and the differential operator is now $\left(\partial^2\eta_{\mu\nu} - \partial_\mu\partial_\nu\right)$. Its solution is given by (cf. (70)):

$$Z[J] = \mathcal{N}\exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x})(\hat{M}^{-1})^{\mu\nu}J_\nu(\boldsymbol{y})\, d^4x\, d^4y\right) \tag{83}$$

$$\text{with } \mathcal{N} = \lim_{N\to\infty}\sqrt{\frac{(2\pi i)^{N-1}}{\det(\hat{M})}}$$

where $\hat{M} = \partial^2\eta_{\mu\nu} - \partial_\mu\partial_\nu$ is the electromagnetic field operator. However, we are in serious trouble because the determinant of $\hat{M}$ is zero and therefore its inverse $\hat{M}^{-1}$ does not exist.

5.2 The issue of gauge invariance

We can easily show that $\hat{M}$ is not invertible since the eigenvalue associated with $\partial_\nu \alpha(\boldsymbol{x})$ is zero:

$$\hat{M}^{\mu\nu}(\partial_\nu \alpha(\boldsymbol{x})) = (\partial^2 \eta^{\mu\nu} - \partial^\mu \partial^\nu)\partial_\nu \alpha(\boldsymbol{x})$$

$$\hat{M}^{\mu\nu}(\partial_\nu \alpha(\boldsymbol{x})) = \partial^2 \eta^{\mu\nu} \partial_\nu \alpha(\boldsymbol{x}) - \partial^\mu \partial^\nu \partial_\nu \alpha(\boldsymbol{x})$$

$$\hat{M}^{\mu\nu}(\partial_\nu \alpha(\boldsymbol{x})) = \partial^2 \partial^\mu \alpha(\boldsymbol{x}) - \partial^\mu \partial^2 \alpha(\boldsymbol{x}) = 0$$

■

The non-invertibility, or *singularity*, of $\hat{M}$, stems from the *gauge invariance* of electromagnetism. Any gauge transformation $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) + \partial^\mu \alpha(\boldsymbol{x})$, with $\alpha(\boldsymbol{x})$ an arbitrary scalar field, leaves the electromagnetic field tensor $F^{\mu\nu} = \partial^\mu A^\nu(\boldsymbol{x}) - \partial^\nu A^\mu(\boldsymbol{x})$ unchanged:

$$\begin{aligned}
\partial^\mu A^\nu(\boldsymbol{x}) - \partial^\nu A^\mu(\boldsymbol{x}) \;&\to\; \partial^\mu (A^\nu(\boldsymbol{x}) + \partial^\nu \alpha(\boldsymbol{x})) - \partial^\nu (A^\mu(\boldsymbol{x}) + \partial^\mu \alpha(\boldsymbol{x})) \\
&= \partial^\mu A^\nu(\boldsymbol{x}) + \partial^\mu \partial^\nu \alpha(\boldsymbol{x}) - \partial^\nu A^\mu(\boldsymbol{x}) - \partial^\nu \partial^\mu \alpha(\boldsymbol{x}) \\
&= \partial^\mu A^\nu(\boldsymbol{x}) - \partial^\nu A^\mu(\boldsymbol{x}) = F^{\mu\nu}
\end{aligned}$$

This gauge invariance means that there exists an infinite number of $A_\mu(\boldsymbol{x})$ variations that result in the same physical electromagnetic field configuration. All these redundant variations of $A_\mu(\boldsymbol{x})$ accumulate in the path integral (82)

$$Z[J] = \int \mathcal{D}[A^\mu(\boldsymbol{x})] \exp\left(\frac{i}{2} \int A^\mu(\boldsymbol{x}) \left(\partial^2 \eta_{\mu\nu} - \partial_\mu \partial_\nu \right) A^\nu(\boldsymbol{x})\, d^4x + i \int J_\mu(\boldsymbol{x}) A^\mu(\boldsymbol{x}) d^4x \right)$$

and eventually make $Z[J]$ diverge to infinity. This type of *singularity-induced* divergence is easier to comprehend with the following n-dimensional integral:

$$\int_{-\infty}^{+\infty} dx_1 \cdots dx_n \, e^{-\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}} \tag{84}$$

where $\mathbf{x}$ is an n-dimensional vector, and $\boldsymbol{A}$ is an $n \times n$ matrix. Suppose that one of the eigenvalues of $\boldsymbol{A}$ is 0, while the $n-1$ others are all equal to 1:

$$\boldsymbol{A} = \begin{pmatrix} 1 & & & \\ & 0 & & \\ & & 1 & \\ & & & \ddots \end{pmatrix} \quad \Rightarrow \quad \mathbf{x}^T \boldsymbol{A}\, \mathbf{x} = x_1^2 + 0 + x_3^2 + \cdots + x_n^2$$

With this kind of singular matrix, the integral in (84) evaluates to:

$$\int_{-\infty}^{+\infty} dx_1 \cdots dx_n e^{-\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}} = \int_{-\infty}^{+\infty} dx_1 \cdots dx_n \, e^{-\left(x_1^2 + x_3^2 + \cdots + x_n^2\right)}$$

$$\int_{-\infty}^{+\infty} dx_1 \cdots dx_n e^{-\mathbf{x}^T \boldsymbol{A}\, \mathbf{x}} = \int_{-\infty}^{+\infty} dx_2 \int_{-\infty}^{+\infty} dx_1 dx_3 \cdots dx_n \, e^{-\left(x_1^2 + x_3^2 + \cdots + x_n^2\right)}$$

Just like the integral over dx_2 diverges badly, the infinite variations of $A_\mu(\boldsymbol{x})$ cause the path integral in (82) diverge. We need to find a way to get rid of all these redundant field configurations, and retain only a single version of $A_\mu(\boldsymbol{x})$ that acts as the designated representative for all the other $A_\mu(\boldsymbol{x}) + \partial_\mu \alpha(\boldsymbol{x})$ variations that lead to the same physical configuration of the electromagnetic field. Because we are dealing with an integral, the idea first introduced by Faddeev and Popov is to inject a delta function that picks out one configuration of $A_\mu(\boldsymbol{x})$ and returns zero for its $A_\mu(\boldsymbol{x}) + \partial_\mu \alpha(\boldsymbol{x})$ variations [12].

5.3 The Faddeev-Popov identity

To begin, let us restate the integral change of variables formula [13]:

$$\int f(x_1,\ldots,x_n)\,dx_1\cdots dx_n = \int f\left(g_1(y_1,\ldots,y_n),\cdots,g_n(y_1,\ldots,y_n)\right)\left|\det(\boldsymbol{G})\right|\,dy_1\cdots dy_n \quad (85)$$

where the function $g_i(y_1,\cdots,y_n)$ expresses the original variable x_i in terms of the new variables $(y_1,\cdots,y_n)$. The determinant $|\det(\boldsymbol{G})|$ is the *Jacobian*, and $\boldsymbol{G}$ is the *Jacobian matrix* associated with the $(x_1,\cdots,x_n)\to(y_1,\cdots,y_n)$ change of variables:

$$\boldsymbol{G} = \begin{pmatrix} \frac{\partial g_1}{\partial y_1} & \cdots & \frac{\partial g_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_n}{\partial y_1} & \cdots & \frac{\partial g_n}{\partial y_n} \end{pmatrix}$$

Equation (85) is sometimes written

$$\int f(x_1,\ldots,x_n)\,dx_1\cdots dx_n = \int f\left(g_1(y_1,\ldots,y_n),\cdots,g_n(y_1,\ldots,y_n)\right)\left|\det\left(\frac{\partial g_i}{\partial y_j}\right)\right|\,dy_1\cdots dy_n$$

to explicitly show what the (i,j) component of $\boldsymbol{G}$ represents. With this refresher in mind, let us derive the Faddeev-Popov identity. The fundamental property of the n-dimensional delta function is that:

$$\int \delta^{(n)}(x_1,\ldots,x_n)\,dx_1\cdots dx_n = 1 \Leftrightarrow \int \delta(x_1)\times\cdots\times\delta(x_n)\,dx_1\cdots dx_n = 1$$

Applying the change of variables $(x_1,\cdots,x_n)\to(y_1,\cdots,y_n)$, with $x_i = g_i(y_1,\ldots,y_n)$, to this delta function property yields:

$$\int \delta(g_1(y_1,\ldots,y_n))\times\cdots\times\delta(g_n(y_1,\ldots,y_n))\left|\det\left(\frac{\partial g_i}{\partial y_j}\right)\right|\,dy_1\cdots dy_n = 1 \quad (86)$$

Before we can use this delta function inside the path integral (83), we must generalize it to a *delta functional*. The transformation proceeds as follows:

1. The discrete index i in y_i becomes a continuous spacetime variable, while the n integration variables become a continuous function of spacetime $\alpha(\boldsymbol{x})$:

$$(y_1,\ldots,y_n)\to\alpha(\boldsymbol{x})$$

2. The classical integral over the dy_i variables becomes a path integral over all possible configurations of the α function:

$$\int dy_1\cdots dy_n \to \int \mathcal{D}[\alpha]$$

3. Each function $g_i(y_1,\ldots,y_n)$ becomes a functional of α:

$$g_i(y_1,\ldots,y_n)\to G_i[\alpha]$$

4. The set of n functions $g_i(y_1,\ldots,y_n)$ becomes a *functional operator*:

$$\left(g_1(y_1,\ldots,y_n),\cdots,g_n(y_1,\ldots,y_n)\right)\to G[\alpha](\boldsymbol{x})$$

   G takes a spacetime function α and returns another spacetime function $G[\alpha](\boldsymbol{x})$.

5. The classical derivative ∂ becomes a functional derivative:

$$\frac{\partial g_i}{\partial y_i} \to \frac{\delta G[\alpha](\boldsymbol{x})}{\delta \alpha(\boldsymbol{y})}$$

The functional derivative $\delta G[\alpha](\boldsymbol{x})/\delta\alpha(\boldsymbol{y})$ represents the infinitesimal variation of the scalar $G[\alpha](\boldsymbol{x})$ that is induced by an infinitesimal variation of the function α at $\boldsymbol{y}$.

6. The product of the n one-dimensional delta functions becomes an infinite product of delta functions; one for each spacetime point:

$$\delta(g_1(y_1,\ldots,y_n)) \times \cdots \times \delta(g_n(y_1,\ldots,y_n)) \to \prod_{\boldsymbol{x}\in\mathbb{M}} \delta\big[G[\alpha](\boldsymbol{x})\big] := \delta\big[G[\alpha]\big]$$

Said differently, the delta functional $\delta\big[G[\alpha]\big]$ is zero everywhere, except if the function $G[\alpha](\boldsymbol{x})$ is zero everywhere.

After applying these six steps to equation (86), one finds the Faddeev-Popov identity:

$$\int \mathcal{D}[\alpha]\, \delta\big[G[\alpha]\big] \left|\det\left(\frac{\delta G[\alpha]}{\delta\alpha}\right)\right| = 1 \tag{87}$$

The term $\left|\det\left(\frac{\delta G[\alpha]}{\delta\alpha}\right)\right|$ is the *Faddeev-Popov determinant*, and it can be seen as the scale factor necessary to normalize the integral to 1. While this result holds for any functional $G[\alpha]$, the trick is to choose one that eliminates the redundancies causing the divergence.

5.4 Gauge fixing of the generating functional

Because the right-hand side of (87) is equal to 1 (hence the name Faddeev-Popov *identity*), we can insert the entire expression on the left-hand side into our original expression for $Z[J]$ without changing the result.

$$Z[J] = \int \mathcal{D}[\alpha] \int \mathcal{D}[A^\mu(\boldsymbol{x})]\delta\big[G[\alpha]\big] \left|\det\left(\frac{\delta G[\alpha]}{\delta\alpha}\right)\right| \exp\left(i\int \left(\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})\right) d^4x\right) \tag{88}$$

We choose G to be the functional operator defined by

$$G[\alpha](\boldsymbol{x}) = \partial^\mu(A_\mu(\boldsymbol{x}) + \partial_\mu\alpha(\boldsymbol{x})) - \omega(\boldsymbol{x}) \tag{89}$$

where $\omega(\boldsymbol{x})$ is an arbitrary scalar function. Again, G is a functional operator that takes a function $\alpha(\boldsymbol{x})$ and returns the function $\partial^\mu(A_\mu(\boldsymbol{x}) + \partial_\mu\alpha(\boldsymbol{x})) - \omega(\boldsymbol{x})$, and where the electromagnetic 4-potential $A_\mu(\boldsymbol{x})$ merely acts as a *parameter* of G. Let us calculate the Faddeev-Popov determinant associated with this functional operator G:

$$\frac{\delta G[\alpha](\boldsymbol{x})}{\delta\alpha(\boldsymbol{y})} = \frac{\delta}{\delta\alpha(\boldsymbol{y})}\Big(\partial^\mu A_\mu(\boldsymbol{x}) + \partial^2\alpha(\boldsymbol{x}) - \omega(\boldsymbol{x})\Big)$$

$$\frac{\delta G[\alpha](\boldsymbol{x})}{\delta\alpha(\boldsymbol{y})} = \frac{\delta}{\delta\alpha(\boldsymbol{y})}\Big(\partial^2\alpha(\boldsymbol{x})\Big) = \partial^2 \times \frac{\delta\alpha(\boldsymbol{x})}{\delta\alpha(\boldsymbol{y})}$$

$$\frac{\delta G[\alpha](\boldsymbol{x})}{\delta\alpha(\boldsymbol{y})} = \partial^2\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$$

$$\det\left(\frac{\delta G[\alpha]}{\delta\alpha}\right) = \det\Big(\partial^2\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})\Big) = \det\big(\partial^2\big)\det\Big(\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})\Big) \tag{90}$$

The 4-dimensional delta function $\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$ is essentially the equivalent of the identity matrix for linear operators. Specifically, it represents the $(\boldsymbol{x},\boldsymbol{y})$ component of the identity operator written as a matrix in the position eigenbasis:

$$\langle \boldsymbol{x}|\mathbb{I}|\boldsymbol{y}\rangle = \langle \boldsymbol{x}|\boldsymbol{y}\rangle = \delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$$

Another way to see this it to recall that the determinant is the product of the operator's eigenvalues. The eigenvalue equation for a discrete linear operator $\boldsymbol{L}$ is:

$$\boldsymbol{L}\boldsymbol{a} = \lambda \boldsymbol{a} \quad \Leftrightarrow \quad \sum_j L_{ij}a_j = \lambda a_i$$

The generalization to a continuous operator consists of substituting the matrix $\boldsymbol{L}$ with a continuous operator $\hat{L}$, the vector $\boldsymbol{a}$ with a continuous function $\phi(\boldsymbol{x})$, and the discrete sum $\sum_j$ with a continuous integral $\int d^4y$:

$$\hat{L}\phi(\boldsymbol{x}) = \lambda\phi(\boldsymbol{x}) \quad \Leftrightarrow \quad \int L(\boldsymbol{x},\boldsymbol{y})\phi(\boldsymbol{y})d^4y = \lambda\phi(\boldsymbol{x})$$

Applying this continuous eigenvalue equation for the operator $\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$ yields:

$$\int \delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})\phi(\boldsymbol{y})d^4y = \lambda\phi(\boldsymbol{x})$$

$$\phi(\boldsymbol{x}) = \lambda\phi(\boldsymbol{x}) \Rightarrow \lambda = 1$$

All the eigenvalues of $\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$ are equal to 1, which means that its determinant is also equal to 1. The Faddeev-Popov determinant in (90) thus simplifies to:

$$\det\left(\frac{\delta G[\alpha]}{\delta\alpha}\right) = \det\left(\partial^2\right) \tag{91}$$

Since the Faddeev-Popov determinant does not depend on α or A^μ, we can factor it out of the integrals in (88) and write:

$$Z[J] = \det\left(\partial^2\right)\int \mathcal{D}[\alpha]\int \mathcal{D}[A^\mu(\boldsymbol{x})]\delta\big[G[\alpha]\big]\exp\left(i\int\left(\mathcal{L}+J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})\right)d^4x\right) \tag{92}$$

The next step is to implement the change of variables $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})$. Let us study its impact on the various terms inside (92). The integration measure $\mathcal{D}[A_\mu(\boldsymbol{x})]$ is obviously unaffected by $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})$, since it merely shifts $A^\mu(\boldsymbol{x})$. We have already seen that gauge transformations like $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})$ leave the electromagnetic field tensor $F^{\mu\nu}$ unchanged (cf. 5.2), and therefore the Lagrangian $\mathcal{L} = -F_{\mu\nu}F^{\mu\nu}/4$ is also unchanged. Next we show that the term $\int J_\mu A^\mu d^4x$ is unaffected by this change of variables:

$$\begin{aligned}\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})d^4x \;\to\; &\int J_\mu(\boldsymbol{x})\left(A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})\right)d^4x \\ &= \int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})d^4x - \int J_\mu(\boldsymbol{x})\partial^\mu\alpha(\boldsymbol{x})d^4x\end{aligned}$$

Integrating by parts the last integral, one finds:

$$\int J_\mu(\boldsymbol{x})\partial^\mu\alpha(\boldsymbol{x})d^4x = \Big[J_\mu(\boldsymbol{x})\alpha(\boldsymbol{x})\Big]_{-\infty}^{+\infty} - \int \alpha(\boldsymbol{x})\partial^\mu J_\mu(\boldsymbol{x})d^4x$$

The boundary term is zero because we are assuming that there is no charge current at the boundaries of the spacetime integration volume, i.e. that $J_\mu(\boldsymbol{x}) = 0$ for $|\boldsymbol{x}| \to \infty$. Similarly,

the conservation of the electric current $\partial^\mu J_\mu(\boldsymbol{x}) = 0$ means that the last term is also equal to zero. This proves that $\int J_\mu A^\mu d^4x$ is indeed invariant under the change of variables $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})$. We now turn to the last term in (92): the delta functional $\delta\big[G[\alpha]\big]$.

$$\begin{aligned}\delta\big[G[\alpha]\big] = \delta\big[\partial^\mu(A_\mu(\boldsymbol{x}) + \partial_\mu\alpha(\boldsymbol{x})) - \omega(\boldsymbol{x})\big] \;&\to\; \delta\big[\partial^\mu(A_\mu(\boldsymbol{x}) - \partial_\mu\alpha(\boldsymbol{x}) + \partial_\mu\alpha(\boldsymbol{x})) - \omega(\boldsymbol{x})\big] \\ &= \delta\big[\partial^\mu A_\mu(\boldsymbol{x}) - \omega(\boldsymbol{x})\big]\end{aligned}$$

In summary, the change of variables $A^\mu(\boldsymbol{x}) \to A^\mu(\boldsymbol{x}) - \partial^\mu\alpha(\boldsymbol{x})$ in equation (92) yields:

$$Z[J] = \det\big(\partial^2\big)\int \mathcal{D}[\alpha]\int \mathcal{D}[A^\mu(\boldsymbol{x})]\delta\big[\partial^\mu A_\mu(\boldsymbol{x}) - \omega(\boldsymbol{x})\big]\exp\left(i\int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\,d^4x\right) \quad (93)$$

The definition (89) of G includes an arbitrary scalar function $\omega(\boldsymbol{x})$ that we can choose to have the following property:

$$\mathcal{K}\int \mathcal{D}[\omega(\boldsymbol{x})]\exp\left(-\frac{i}{2\xi}\int \omega(\boldsymbol{x})^2 d^4x\right) = 1 \quad (94)$$

where ξ is a scalar *gauge parameter*, and $\mathcal{K}$ is the constant chosen such that this functional Gaussian integral is normalized to 1. Because (94) is equal to 1, we can insert the left-hand side into (93) to find:

$$\begin{aligned}Z[J] = \det\big(\partial^2\big)\mathcal{K}\int \mathcal{D}[\omega(\boldsymbol{x})]\int \mathcal{D}[\alpha]\int \mathcal{D}[A^\mu(\boldsymbol{x})] \\ \exp\left(-\frac{i}{2\xi}\int \omega(\boldsymbol{x})^2 d^4x\right)\delta\big[\partial^\mu A_\mu(\boldsymbol{x}) - \omega(\boldsymbol{x})\big]\exp\left(i\int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\,d^4x\right)\end{aligned}$$

Thanks to the delta functional $\delta\big[\partial^\mu A_\mu(\boldsymbol{x}) - \omega(\boldsymbol{x})\big]$, we can replace $\omega(\boldsymbol{x})$ with $\partial^\mu A_\mu(\boldsymbol{x})$ and remove the path integral over $\mathcal{D}[\omega]$:

$$\begin{aligned}Z[J] = \det\big(\partial^2\big)\mathcal{K}\int \mathcal{D}[\alpha]\int \mathcal{D}[A^\mu(\boldsymbol{x})] \\ \exp\left(-\frac{i}{2\xi}\int (\partial^\mu A_\mu(\boldsymbol{x}))^2 d^4x\right)\exp\left(i\int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\,d^4x\right)\end{aligned}$$

After integrating by parts $\int(\partial^\mu A_\mu(\boldsymbol{x}))^2 d^4x$, and again assuming that the field vanishes at the boundary of the spacetime integration volume, one finds:

$$\int (\partial^\mu A_\mu(\boldsymbol{x}))^2\,d^4x = \int (\partial^\mu A_\mu(\boldsymbol{x}))(\partial^\nu A_\nu(\boldsymbol{x}))\,d^4x$$

$$\int (\partial^\mu A_\mu(\boldsymbol{x}))^2\,d^4x = \Big[A_\mu(\boldsymbol{x})\partial^\nu A_\nu(\boldsymbol{x})\Big]_{-\infty}^{+\infty} - \int A_\mu(\boldsymbol{x})(\partial^\mu\partial^\nu A_\nu(\boldsymbol{x}))\,d^4x$$

$$\int (\partial^\mu A_\mu(\boldsymbol{x}))^2\,d^4x = -\int A_\mu(\boldsymbol{x})\partial^\mu\partial^\nu A_\nu(\boldsymbol{x})\,d^4x = -\int A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})\,d^4x$$

Our expression for $Z[J]$ thus becomes:

$$\begin{aligned}Z[J] = \det\big(\partial^2\big)\mathcal{K}\int \mathcal{D}[\alpha]\int \mathcal{D}[A^\mu(\boldsymbol{x})] \\ \exp\left(\frac{i}{2\xi}\int A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})d^4x\right)\exp\left(i\int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\,d^4x\right)\end{aligned} \quad (95)$$

Recalling that in section 5.1 we found that

$$i\int (\mathcal{L} + J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x}))\,d^4x$$

was equivalent to (cf. (82))

$$\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu}-\partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\,d^4x+i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})d^4x$$

the expression for $Z[J]$ in equation (95) becomes:

$$Z[J]=\det\left(\partial^2\right)\mathcal{K}\int\mathcal{D}[\alpha]\int\mathcal{D}[A^\mu(\boldsymbol{x})]\exp\left(\frac{i}{2\xi}\int A^\mu(\boldsymbol{x})\partial_\mu\partial_\nu A^\nu(\boldsymbol{x})d^4x\right)$$
$$\times\exp\left(\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu}-\partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\,d^4x+i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})d^4x\right)$$

$$Z[J]=\det\left(\partial^2\right)\mathcal{K}\int\mathcal{D}[\alpha]\int\mathcal{D}[A^\mu(\boldsymbol{x})]$$
$$\times\exp\left(\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu}-\partial_\mu\partial_\nu+\frac{\partial_\mu\partial_\nu}{\xi}\right)A^\nu(\boldsymbol{x})\,d^4x+i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})\,d^4x\right)$$

$$Z[J]=\det\left(\partial^2\right)\mathcal{K}\int\mathcal{D}[\alpha]$$
$$\int\mathcal{D}[A^\mu(\boldsymbol{x})]\exp\left(\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu}-\left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\,d^4x+i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})\,d^4x\right)$$

The path integral over $\mathcal{D}[A^\mu(\boldsymbol{x})]$ is a functional Gaussian integral whose solution was derived in section 4.2. As applied to the path integral above, we find:

$$\int\mathcal{D}[A^\mu(\boldsymbol{x})]\exp\left(\frac{i}{2}\int A^\mu(\boldsymbol{x})\left(\partial^2\eta_{\mu\nu}-\left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\right)A^\nu(\boldsymbol{x})\,d^4x+i\int J_\mu(\boldsymbol{x})A^\mu(\boldsymbol{x})\,d^4x\right)$$
$$=\exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x})(\hat{M}^{-1})^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})J_\nu(\boldsymbol{y})\,d^4x\,d^4y\right)$$

$$\Rightarrow Z[J]=\left[\det\left(\partial^2\right)\mathcal{K}\int\mathcal{D}[\alpha]\right]\exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x})(\hat{M}^{-1})^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})J_\nu(\boldsymbol{y})\,d^4x\,d^4y\right)\qquad(96)$$

where $(\hat{M}^{-1})^{\mu\nu}$ is the (μ,ν) component of the inverse of operator $\hat{M}^{\mu\nu}=\partial^2\eta^{\mu\nu}-(1-1/\xi)\,\partial^\mu\partial^\nu$. Unlike the non-invertible operator $\left(\partial^2\eta^{\mu\nu}-\partial^\mu\partial^\nu\right)$ we found in section 5.1, this new operator is invertible thanks to the gauge-fixing factor $(1-1/\xi)$. This justifies the seemingly strange choice we made for $\omega(\boldsymbol{x})$ in (94). To prove that $\hat{M}^{\mu\nu}$ is invertible, let us find its Fourier transform in momentum space (the $1/2\pi$ factors are omitted for clarity):

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p})=\int\left[\hat{M}^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})A_\nu(\boldsymbol{x})\right]e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}\,d^4x$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p})=\int\left[\left(\partial^2\eta^{\mu\nu}-\left(1-\frac{1}{\xi}\right)\partial^\mu\partial^\nu\right)A_\nu(\boldsymbol{x})\right]e^{-ip^\mu x_\mu}\,d^4x$$

Integrating by parts, and discarding the boundary term because the field vanishes at the space-time boundaries, we find[3]:

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p})=\int\left[\left(\partial^2\eta^{\mu\nu}-\left(1-\frac{1}{\xi}\right)\partial^\mu\partial^\nu\right)e^{-ip^\mu x_\mu}\right]A_\nu(\boldsymbol{x})\,d^4x$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p})=\int\left[\left((-ip^\sigma)(-ip_\sigma)\eta^{\mu\nu}-\left(1-\frac{1}{\xi}\right)(-ip^\mu)(-ip^\nu)\right)e^{-ip^\mu x_\mu}\right]A_\nu(\boldsymbol{x})\,d^4x$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p})=\int\left(-\boldsymbol{p}^2\eta^{\mu\nu}+\left(1-\frac{1}{\xi}\right)p^\mu p^\nu\right)e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}A_\nu(\boldsymbol{x})\,d^4x$$

[3]Note that there is no sign inversion because we are integrating by parts twice: once for each second derivative ∂^2 and $\partial^\mu\partial^\nu$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p}) = \left(-\boldsymbol{p}^2\eta^{\mu\nu} + \left(1-\frac{1}{\xi}\right)p^\mu p^\nu\right)\int e^{-i\boldsymbol{p}\cdot\boldsymbol{x}}A_\nu(\boldsymbol{x})\,d^4x$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})A_\nu(\boldsymbol{p}) = \left(-\boldsymbol{p}^2\eta^{\mu\nu} + \left(1-\frac{1}{\xi}\right)p^\mu p^\nu\right)A_\nu(\boldsymbol{p})$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p}) = -\boldsymbol{p}^2\eta^{\mu\nu} + \left(1-\frac{1}{\xi}\right)p^\mu p^\nu \tag{97}$$

With $\hat{M}$ written in momentum space, it is now much easier to find that it does have an inverse:

$$\hat{M}^{\mu\sigma}(\boldsymbol{p})\left(-\frac{\eta_{\sigma\nu}}{\boldsymbol{p}^2} + (1-\xi)\frac{p_\sigma p_\nu}{\boldsymbol{p}^4}\right) = \left(-\boldsymbol{p}^2\eta^{\mu\sigma} + \left(1-\frac{1}{\xi}\right)p^\mu p^\sigma\right)\left(-\frac{\eta_{\sigma\nu}}{\boldsymbol{p}^2} + (1-\xi)\frac{p_\sigma p_\nu}{\boldsymbol{p}^4}\right)$$

$$= \eta^{\mu\sigma}\eta_{\sigma\nu} - (1-\xi)\frac{\eta^{\mu\sigma}p_\sigma p_\nu}{\boldsymbol{p}^2} - \left(1-\frac{1}{\xi}\right)\frac{p^\mu p^\sigma \eta_{\sigma\nu}}{\boldsymbol{p}^2} + \left(1-\frac{1}{\xi}\right)(1-\xi)\frac{p^\mu p^\sigma p_\sigma p_\nu}{\boldsymbol{p}^4}$$

$$= \delta^\mu_\nu - (1-\xi)\frac{p^\mu p_\nu}{\boldsymbol{p}^2} - \left(1-\frac{1}{\xi}\right)\frac{p^\mu p_\nu}{\boldsymbol{p}^2} + \left(1-\frac{1}{\xi}\right)\frac{p^\mu \boldsymbol{p}^2 p_\nu}{\boldsymbol{p}^4} + (1-\xi)\frac{p^\mu \boldsymbol{p}^2 p_\nu}{\boldsymbol{p}^4}$$

$$\hat{M}^{\mu\sigma}(\boldsymbol{p})\left(-\frac{\eta_{\sigma\nu}}{\boldsymbol{p}^2} + (1-\xi)\frac{p_\sigma p_\nu}{\boldsymbol{p}^4}\right) = \delta^\mu_\nu$$

$$\Rightarrow (\hat{M}^{-1})_{\sigma\nu}(\boldsymbol{p}) = \frac{1}{\boldsymbol{p}^2}\left(-\eta_{\sigma\nu} + (1-\xi)\frac{p_\sigma p_\nu}{\boldsymbol{p}^2}\right) \Leftrightarrow (\hat{M}^{-1}(\boldsymbol{p}))^{\mu\nu} = \frac{1}{\boldsymbol{p}^2}\left(-\eta^{\mu\nu} + (1-\xi)\frac{p^\mu p^\nu}{\boldsymbol{p}^2}\right)$$

This proves that our new operator $\hat{M}^{\mu\nu} = \partial^2\eta^{\mu\nu} - (1-1/\xi)\,\partial^\mu\partial^\nu$ is indeed invertible. Incidentally, without the gauge-fixing parameter ξ equation (97) becomes

$$\hat{M}^{\mu\nu}(\boldsymbol{p}) = -\boldsymbol{p}^2\eta^{\mu\nu} + p^\mu p^\nu$$

which has an obvious zero eigenvalue for p_ν:

$$\hat{M}^{\mu\nu}(\boldsymbol{p})\,p_\nu = \left(-\boldsymbol{p}^2\eta^{\mu\nu} + p^\mu p^\nu\right)p_\nu$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})\,p_\nu = -\boldsymbol{p}^2\eta^{\mu\nu}p_\nu + p^\mu p^\nu p_\nu$$

$$\hat{M}^{\mu\nu}(\boldsymbol{p})\,p_\nu = -\boldsymbol{p}^2 p^\mu + p^\mu\boldsymbol{p}^2 = 0$$

This provides a definite mathematical justification as to why the ξ gauge-fixing parameter removes the singularity of the electromagnetic field operator. Writing $G^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})$ for the (μ,ν) component of the Green function associated with the gauge-fixed electromagnetic field, equation (96) becomes:

$$Z[J] = \left[\det\left(\partial^2\right)\mathcal{K}\int\mathcal{D}[\alpha]\right]\exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x})G^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})J_\nu(\boldsymbol{y})\,d^4x\,d^4y\right) \tag{98}$$

Note that, unlike the Klein-Gordon field scalar Green function we derived in section 4.2, the electromagnetic field Green function is a second-rank tensor. For a scalar field, the Green function $G(\boldsymbol{x},\boldsymbol{y})$ describes how the field value at $\boldsymbol{y}$ responds to an impulse at $\boldsymbol{x}$. For the 4-dimensional electromagnetic vector field A^μ, the Green function component $G^{\mu\nu}(\boldsymbol{x},\boldsymbol{y})$ relates the A^ν component of the field at spacetime point $\boldsymbol{y}$ with the A^μ component of the field at $\boldsymbol{x}$. Because of its tensor form, the electromagnetic Green function inverse equation is slightly different from the scalar case and is given by:

$$\left(\partial^2\eta_{\mu\nu} - (1-1/\xi)\,\partial_\mu\partial_\nu\right)G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \delta^\sigma_\mu\delta^{(4)}(\boldsymbol{x}-\boldsymbol{y}) \tag{99}$$

Finally, the prefactor $\left[\det(\partial^2)\mathcal{K}\int\mathcal{D}[\alpha]\right]$ in (98) is yet another infinite constant that will be factored out during normalization.

5.5 The Feynman propagator of the electromagnetic field

To show that the gauge-fixed electromagnetic Green function is given by

$$G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{(2\pi)^4}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{\boldsymbol{p}^\sigma\boldsymbol{p}^\nu}{\boldsymbol{p}^2}\right)d^4p \tag{100}$$

we have to prove that it verifies the Green function inverse equation (99):

$$\left(\partial^2\eta_{\mu\nu}-\left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\right)G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{(2\pi)^4}\left[\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p \right.$$
$$\left. -\left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p\right]$$

We will perform the detailed calculation of the two terms in the bracket separately.

Term 1

$$\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p = \eta_{\mu\nu}\int \frac{\partial^2(e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}})}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p = \eta_{\mu\nu}\int \frac{-\boldsymbol{p}^2}{\boldsymbol{p}^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p = -\eta_{\mu\nu}\int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p = \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\left(\eta_{\mu\nu}\eta^{\sigma\nu}-(1-\xi)\frac{p^\sigma\eta_{\mu\nu}p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$\partial^2\eta_{\mu\nu}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p = \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\left(\delta^\sigma_\mu-(1-\xi)\frac{p^\sigma p_\mu}{\boldsymbol{p}^2}\right)d^4p$$

Term 2

$$\left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$
$$= \left(1-\frac{1}{\xi}\right)\int \frac{\partial_\mu\partial_\nu(e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}})}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1-\frac{1}{\xi}\right)\partial_\mu\partial_\nu\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1-\frac{1}{\xi}\right)\int \frac{\partial_\mu\partial_\nu(e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}})}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1-\frac{1}{\xi}\right)\int \frac{\partial_\mu(-ip_\nu e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}})}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1-\frac{1}{\xi}\right)\int \frac{(-ip_\mu)(-ip_\nu)e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1-\frac{1}{\xi}\right)\int \frac{-p_\mu p_\nu}{\boldsymbol{p}^2}e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right)d^4p$$

$$= \left(1 - \frac{1}{\xi}\right) \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\frac{p_\mu p_\nu \eta^{\sigma\nu}}{\boldsymbol{p}^2} - \frac{p_\mu p_\nu}{\boldsymbol{p}^2}(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right) d^4p$$

$$= \left(1 - \frac{1}{\xi}\right) \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\frac{p_\mu p^\sigma}{\boldsymbol{p}^2} - \frac{p_\mu}{\boldsymbol{p}^2}(1-\xi)\frac{p^\sigma p_\nu p^\nu}{\boldsymbol{p}^2}\right) d^4p$$

$$= \left(1 - \frac{1}{\xi}\right) \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\frac{p^\sigma p_\mu}{\boldsymbol{p}^2} - \frac{1}{\boldsymbol{p}^2}(1-\xi)\frac{p^\sigma p_\mu \boldsymbol{p}^2}{\boldsymbol{p}^2}\right) d^4p$$

$$= \left(1 - \frac{1}{\xi}\right) \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\frac{p^\sigma p_\mu}{\boldsymbol{p}^2} - \frac{p^\sigma p_\mu}{\boldsymbol{p}^2}(1-\xi)\right) d^4p$$

$$= \left(1 - \frac{1}{\xi}\right) \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \frac{p^\sigma p_\mu}{\boldsymbol{p}^2}\xi \, d^4p$$

$$\Rightarrow \left(1 - \frac{1}{\xi}\right) \partial_\mu\partial_\nu \int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2} \left(-\eta^{\sigma\nu} + (1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2}\right) d^4p = \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \frac{p^\sigma p_\mu}{\boldsymbol{p}^2}(\xi - 1)\, d^4p$$

Substituting these two terms into our Green function inverse equation yields:

$$\left(\partial^2\eta_{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right) G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{(2\pi)^4}\left[\int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\delta^\sigma_\mu - (1-\xi)\frac{p^\sigma p_\mu}{\boldsymbol{p}^2}\right) d^4p \right.$$
$$\left. - \int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \frac{p^\sigma p_\mu}{\boldsymbol{p}^2}(\xi - 1)\, d^4p\right]$$

$$\left(\partial^2\eta_{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right) G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{(2\pi)^4}\left[\int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}} \left(\delta^\sigma_\mu - (1-\xi)\frac{p^\sigma p_\mu}{\boldsymbol{p}^2} - \frac{p^\sigma p_\mu}{\boldsymbol{p}^2}(\xi - 1)\right) d^4p\right]$$

$$\left(\partial^2\eta_{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right) G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \delta^\sigma_\mu \frac{1}{(2\pi)^4}\int e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\, d^4p$$

$$\left(\partial^2\eta_{\mu\nu} - \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right) G^{\sigma\nu}(\boldsymbol{x},\boldsymbol{y}) = \delta^\sigma_\mu \delta^{(4)}(\boldsymbol{x}-\boldsymbol{y})$$

■

This confirms that (100) is the electromagnetic field Green function associated. To find the Feynman propagator, we simply have to follow the contour prescription from section 1.3:

$$G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = \lim_{\varepsilon\to 0^+} \frac{1}{(2\pi)^4}\int \frac{e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2 + i\varepsilon}\left(-\eta^{\sigma\nu} + (1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2 + i\varepsilon}\right) d^4p \tag{101}$$

5.6 The photon two-point correlation function

In quantum electrodynamics, the photon is the gauge boson associated with the electromagnetic gauge field. To find the photon two-point correlation function we go back to equation (60) (note we are still working in natural units where $\hbar = 1$):

$$\Delta^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = \frac{(-i)^2}{Z[0]} \left.\frac{\delta^2 Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})}\right|_{J=0} \tag{102}$$

where the electromagnetic generating functional is given by (cf. (98)):

$$Z[J] = \left[\det\left(\partial^2\right)\mathcal{K}\int \mathcal{D}[\alpha]\right] \exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x}) G_F^{\mu\nu}(\boldsymbol{x}-\boldsymbol{y}) J_\nu(\boldsymbol{y})\, d^4x\, d^4y\right)$$

Starting with the functional derivative with respect to $J_\sigma(\boldsymbol{x})$:

$$\frac{\delta Z[J]}{\delta J_\sigma(\boldsymbol{x})} = \left[\det\left(\partial^2\right)\mathcal{K}\int \mathcal{D}[\alpha]\right]\frac{1}{\delta J_\sigma(\boldsymbol{x})}\left[\exp\left(-\frac{i}{2}\int J_\mu(\boldsymbol{x}')G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}')\,d^4x'\,d^4y'\right)\right]$$

$$\frac{\delta Z[J]}{\delta J_\sigma(\boldsymbol{x})} = Z[J]\,\frac{1}{\delta J_\sigma(\boldsymbol{x})}\left[-\frac{i}{2}\int J_\mu(\boldsymbol{x}')G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}')\,d^4x'\,d^4y'\right]$$

$$\frac{\delta Z[J]}{\delta J_\sigma(\boldsymbol{x})} = \frac{-iZ[J]}{2}\int\left[\frac{\delta J_\mu(\boldsymbol{x}')}{\delta J_\sigma(\boldsymbol{x})}G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}') + J_\mu(\boldsymbol{x}')G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')\frac{\delta J_\nu(\boldsymbol{y}')}{\delta J_\sigma(\boldsymbol{x})}\right]d^4x'd^4y'$$

Before we proceed with the functional derivative, there is a little subtlety regarding the term $\delta J_\mu(\boldsymbol{x}')/\delta J_\sigma(\boldsymbol{x})$ that we should be aware of. Because the source is a 4-vector, this functional derivative is given by:

$$\frac{\delta J_\mu(\boldsymbol{x}')}{\delta J_\sigma(\boldsymbol{x})} = \delta_\mu^\sigma\,\delta^{(4)}(\boldsymbol{x}'-\boldsymbol{x})$$

$$\Rightarrow \frac{\delta J_\mu(\boldsymbol{x}')}{\delta J_\sigma(\boldsymbol{x})}G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}') = \delta_\mu^\sigma\,\delta^{(4)}(\boldsymbol{x}'-\boldsymbol{x})G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}')$$

$$\frac{\delta J_\mu(\boldsymbol{x}')}{\delta J_\sigma(\boldsymbol{x})}G_F^{\mu\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}') = \delta^{(4)}(\boldsymbol{x}'-\boldsymbol{x})G_F^{\sigma\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}')$$

With this subtlety in mind, we can continue the derivation:

$$\frac{\delta Z[J]}{\delta J_\sigma(\boldsymbol{x})} = \frac{-iZ[J]}{2}\int\left[\delta^{(4)}(\boldsymbol{x}'-\boldsymbol{x})G_F^{\sigma\nu}(\boldsymbol{x}'-\boldsymbol{y}')J_\nu(\boldsymbol{y}') + J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{y}')\delta^{(4)}(\boldsymbol{y}'-\boldsymbol{x})\right]d^4x'd^4y'$$

$$\frac{\delta Z[J]}{\delta J_\sigma(\boldsymbol{x})} = \frac{-iZ[J]}{2}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')J_\nu(\boldsymbol{y}')d^4y' + \int J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]$$

We now take the second derivative with respect to $J_\nu(\boldsymbol{y})$:

$$\begin{aligned}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})} &= \frac{-i}{2}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')J_\nu(\boldsymbol{y}')d^4y' + \int J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\\ &\quad + \frac{-iZ[J]}{2}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')\frac{\delta J_\nu(\boldsymbol{y}')}{\delta J_\nu(\boldsymbol{y})}d^4y' + \int \frac{\delta J_\mu(\boldsymbol{x}')}{\delta J_\nu(\boldsymbol{y})}G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\end{aligned}$$

$$\begin{aligned}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})} &= \frac{-i}{2}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')J_\nu(\boldsymbol{y}')d^4y' + \int J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\\ &\quad + \frac{-iZ[J]}{2}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')\delta^{(4)}(\boldsymbol{y}'-\boldsymbol{y})d^4y' + \int \delta^{(4)}(\boldsymbol{x}'-\boldsymbol{y})G_F^{\nu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\end{aligned}$$

$$\begin{aligned}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})} &= \frac{-i}{2}\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')J_\nu(\boldsymbol{y}')d^4y' + \int J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\\ &\quad + \frac{-iZ[J]}{2}\left[G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) + G_F^{\nu\sigma}(\boldsymbol{y}-\boldsymbol{x})\right]\end{aligned}$$

Evaluating this expression for $J=0$ yields:

$$\begin{aligned}\left.\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})}\right|_{J=0} &= \frac{-i}{2}\frac{\delta Z[0]}{\delta J_\nu(\boldsymbol{y})}\left[\int G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}')J_\nu(\boldsymbol{y}')d^4y' + \int J_\mu(\boldsymbol{x}')G_F^{\mu\sigma}(\boldsymbol{x}'-\boldsymbol{x})d^4x'\right]\\ &\quad + \frac{-iZ[0]}{2}\left[G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) + G_F^{\nu\sigma}(\boldsymbol{y}-\boldsymbol{x})\right]\end{aligned}$$

The functional derivative $\frac{\delta Z[0]}{\delta J_\nu(\boldsymbol{y})}$ must be zero since in the term $Z[0]$ we have set J to zero. At the end of section 1.3 we saw that Feynman propagators are symmetric, and therefore we can substitute $G_F^{\nu\sigma}(\boldsymbol{y}-\boldsymbol{x})$ with $G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y})$. Thanks to these simplifications, our second functional derivative reduces to:

$$\left.\frac{\delta Z[J]}{\delta J_\nu(\boldsymbol{y})\delta J_\sigma(\boldsymbol{x})}\right|_{J=0} = -iZ[0]G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y})$$

We now insert this result into the photon two-point correlation function (102) to find:

$$\Delta^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = \frac{(-i)^2}{Z[0]}\left(-iZ[0]G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y})\right)$$

$$\Delta^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = iG_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y})$$

Finally, replacing $G_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y})$ with the expression we calculated in (101) yields the standard form of the photon two-point correlation function:

$$\Delta^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = \lim_{\varepsilon\to 0^+}\frac{1}{(2\pi)^4}\int \frac{ie^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}}{\boldsymbol{p}^2+i\varepsilon}\left(-\eta^{\sigma\nu}+(1-\xi)\frac{p^\sigma p^\nu}{\boldsymbol{p}^2+i\varepsilon}\right)d^4p \tag{103}$$

Choosing the *Feynman gauge* consists in setting ξ to 1 and yields the result we set out to find at the very beginning of this paper

$$\begin{gathered}\Delta_F^{\sigma\nu}(\boldsymbol{x}-\boldsymbol{y}) = \lim_{\varepsilon\to 0^+}\frac{1}{(2\pi)^4}\int \frac{-i\eta^{\sigma\nu}}{\boldsymbol{p}^2+i\varepsilon}\,e^{i(\boldsymbol{x}-\boldsymbol{y})\cdot\boldsymbol{p}}\,d^4p \\ \Leftrightarrow \\ \Delta_F^{\sigma\nu}(\boldsymbol{p}) = \lim_{\varepsilon\to 0^+}\frac{-i\eta^{\sigma\nu}}{\boldsymbol{p}^2+i\varepsilon}\end{gathered} \tag{104}$$

where the second expression is the Fourier-transformed photon two-point correlation function in momentum space.

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